The observation scheme in the statistics of generalized repunit primes: an exact conditional test of the Lenstra–Pomerance–Wagstaff constant

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Abstract

The Lenstra–Pomerance–Wagstaff (LPW) heuristic predicts $\#\{n \le x : (b^n-1)/(b-1) \text{ prime}\} \sim (e^\gamma/\ln b)\ln x$, tested since 1993 via a "normalized occurrence frequency". Censoring does not bias this frequency: the endpoint form and the OLS slope are unbiased. The real defect: forms of the estimator disagree by up to 29.5%.

We pool $B = 15$ bases ($M = 218$ events). The search frontiers are not recorded; they are Neyman–Scott nuisance parameters, and the profile estimator M/S is inconsistent. If the frontier ignored the finds (condition (A1)), the position D_b of the last find is sufficient for the frontier, and conditionally on it $N_b - 1 \sim \text{Poisson}(\lambda_b D_b)$ exactly. This gives $(M-B)/S = 1.838$, exact 95% interval [1.59, 2.11], $p = 0.67$ against $e^\gamma \approx 1.781$; the old test rejects a true hypothesis in 11–17% of cases. Under (A1) the ten-percent excess over LPW comes mainly from the observation scheme. Without (A1), agreement with LPW persists for any mean gap between the last find and the frontier from 0 to 3.2 mean spacings (2.8 at second order).

LPW at finite n is tested via the heuristic's second-order term: $R_b(p)$ is prime with probability $e^\gamma \ln(a_b p)/(p \ln b)$. Measuring $c_b = \langle \ln a_b \rangle$ from the divisibility of $R_b(p)$ and summing over the primes, under (A1) we obtain 1.78 (1.66–1.78 at every truncation, $p \ge 0.598$), explaining the "slow convergence from below". The homogeneity test ($p = 0.935$) detects only spreads of κ_b with coefficient of variation above 0.37.

Keywords: repunit primes, generalized repunits, Mersenne primes, Lenstra–Pomerance–Wagstaff conjecture, Euler–Mascheroni constant, observation scheme, Type-I censoring, Poisson point process, conditional likelihood, nuisance parameters, Neyman–Scott problem, sufficient statistic, estimator bias, statistical power, computational number theory

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1 Introduction

For an integer base $b \ge 2$ the generalized repunit is

$$R_b(n) = \frac{b^n - 1}{b - 1}, \tag{1}$$

and $R_b(n)$ can be prime only when n is prime. For $b = 2$ these are the Mersenne numbers $M_p = 2^p - 1$. Building on the heuristics of Wagstaff [1] and Pomerance [2], the LPW conjecture asserts that

$$\pi_b(x) = \#\{n \le x : \ R_b(n) \text{ prime}\} \ \sim \ \frac{e^\gamma}{\ln b} \ln x, \tag{2}$$

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where γ is the Euler–Mascheroni constant; a detailed derivation of the heuristic for $b = 2$ is given in the PrimePages reference [3]. The base enters (2) only through the factor $1/\ln b$: the constant $e^{\gamma} \approx 1.7811$ is declared universal. Dubner [4] proposed testing this universality empirically across bases, and Bourdelais [5] has since maintained running estimates of the normalized frequency, publishing them in the comments on OEIS entries [6]. Related questions for the Mersenne sequence alone were discussed recently in [7]. Close in spirit is also [8], which looks for secondary structure in the Mersenne exponents related to the factorization of $p - 1$; methodologically it is built like our §8—stratified conditional likelihood plus a significance test—but it tests for structure *within* a single base, whereas we test for it *between* bases.

An interpretive folklore has grown up around these estimates. Base 13 is described as "lucky" and base 7 as "the unluckiest"; it is noted that the normalized frequencies lie below the conjectured limit $e^{-\gamma} \approx 0.5615$ and converge to it slowly from below. The present work examines these claims, but arrives at conclusions other than those one would naturally expect.

1.1 Contributions

1. **The frequency estimator in common use is not biased.** The scheme in which the fitting interval ends at the largest known index does indeed condition the estimate on the last observation being an event, and the sufficient statistic is distributed as $\mathrm{Gamma}(N, \lambda_b)$ (Proposition 1). But this does *not* imply that the normalized frequency is biased: both the endpoint form $D/(N \ln b)$ and the OLS slope are exactly unbiased under scheme A, and unbiased up to the factor $1 - (1 + \mu)e^{-\mu}$ under scheme B (Proposition 2, Table 2); the plug-in $\mu \approx N_b \geq 7$ puts the deficit at 0.73% for the endpoint form and 2.3% for OLS without the origin (an estimate, not a bound: at the lower 95% bound $\mu = 3.29$ for $N_b = 7$ the deficit is 16.0%). Only the reciprocal $\hat{\kappa} = 1/\hat{G}$ is biased, by the factor $N/(N-1)$ (under scheme A; under scheme B its expectation is infinite), and that is Jensen's inequality, not censoring. Unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another; any "correction" that repairs one of them damages the other (§3).

2. **The genuine defect is instability with respect to the form of the estimator.** Several numerically distinct variants are in circulation. On identical data they disagree by as much as 29.5% (base 18)—1.7 times the median deviation of the estimates from $e^{-\gamma}$ (17.6%), which is what gets interpreted substantively; for two of the fifteen bases the choice of form weighs more than their entire deviation from the LPW prediction. Under the endpoint form base 18 is the third "unluckiest" of fifteen, and under OLS it is the tenth, below the median. The label describes the choice of estimator, not a property of the primes (§3.4). The size of the discrepancy for base 18, however, is created by a single anomalous gap after an origin $n_{\min} = 2$ fixed by arithmetic.

3. **Main result: an exact conditional likelihood for scheme B.** The pivot $\kappa S \sim \mathrm{Gamma}(M, 1)$ is exact under the scheme "observe until the N-th event", where N is fixed by design. The data, however, arise from a fixed frontier: the search covered $(t_0, \ln L_b]$ and the number of finds is random. The unrecorded window widths W_b are nuisance parameters whose number grows with the number of bases; their joint MLE $\hat{W}_b = D_b$ leads to the inconsistent profile estimator M/S—precisely the Neyman–Scott situation [9]. We show (Proposition 3) that D_b is sufficient for W_b when κ is held fixed and that, conditionally on D_b, the quantity $N_b - 1$ is $\mathrm{Poisson}(\lambda_b D_b)$, with *the window width W_b not entering this distribution.* Hence $M - B \mid D \sim \mathrm{Poisson}(\kappa S)$ is an exact Poisson pivot, and $(M - B)/S$ is the conditional maximum-likelihood estimator in the sense of Andersen [10], exactly conditionally unbiased; it is also obtained by integrating over W_b against the measure dW_b (Remark 6). No moment approximations, no exponentially small remainders and no simulation-based inversion of intervals are required (§4). Exactness refers to the Poisson

model (3): if the heuristic is read literally, as Bernoulli trials at the prime indices, the conditional law is Poisson-binomial and the Poisson intervals are conservative or practically exact (§9). The distinction between the schemes is itself classical, and the unbiasedness of $(N-1)/D$ under scheme A and the loss of one observation per stratum are textbook facts; we lay no claim to them. What is new is the setting: the frontier is *undocumented*, so the exposure must be reconstructed from the finds themselves, and the correction turns out to be proportional to the number of bases (Remark 5).

4. **Numerical consequence and its condition.** Under (A1) the exact conditional inference gives $\hat{\kappa} = 1.838$ with a 95% interval [1.594, 2.109] and $p = 0.672$: the excess over LPW shrinks from 10.3% to a statistically insignificant 3.2%. Under scheme B the old Gamma pivot rejects a true hypothesis in 11–17% of cases (§4.4); the spread depends on how far the frontier has advanced, which is precisely why it is reported as a range. Without (A1) the stopping rule enters the inference through a single number—the mean gap m between the last find and the frontier in units of the mean spacing: by Wald's identity $(M - Bm)/S$ is unbiased in the sense of expectations, and with m known the shifted Poisson test approximately keeps its level (0.043–0.056) under every stopping rule considered, partial violations of (A1) included (Table 3). The conclusion "LPW is not rejected" is robust: it holds for every m from 0 (stopping exactly at a find) to 3.2, and to 2.8 with the second-order term (Table 6); this covers drought $g = 2$ as well ($\bar{m} = 2.14$). LPW is not rejected from above for any $m \geq 0$; it would be rejected from below only for a mean gap exceeding 3.2 (2.8) mean spacings. Only the residual excess depends on m: 10.8% at $m = 0$, 3.2% under (A1), and zero at $m = 1.42$.

5. **Key assumption about the frontiers and its cost.** Besides the Poisson model and contiguous coverage of the search (§2), the conclusion that the excess is created by the observation scheme rests on condition (A1): the frontier was chosen without regard to the finds (the condition concerns the actual search horizon, not the published bound; Remark 9). The estimators $(M-1)/S$ and $(M-B)/S$ are built on the same statistics (N_b, D_b) and differ by 6.9%; which of them is right is determined not by the data but by how the data were collected. Without (A1) only an *upper* bound in the sense of expectations holds rigorously: by Wald's identity $\mathbb{E}[M] \geq \kappa\,\mathbb{E}[S]$ under any stopping rule. There is no lower bound: the rule "stop the search after a long stretch without finds" biases even $(M-B)/S$ upward (Remark 10, Table 3); it cannot produce a rejection of LPW from above: the true κ is then lower still, and a rejection from below would require a very long drought (§5.2). An attempt to sidestep the choice by assembling the estimate from per-base pieces that are unbiased by construction fails (§4.6). The four documented frontiers ($b = 2, 10, 22, 26$) do not settle (A1): three of them predate the data snapshot and give only lower bounds on the gap for it, the moment of publication may itself be tied to the finds, and the bases with a published frontier are not a random sample. Only the $b = 2$ frontier is synchronous with the snapshot, and it is consistent with (A1) (§4.5). The only radical remedy is to publish the frontiers L_b.

6. **Lucky bases: what the test cannot do.** Conditioning on $K = M - B$ makes the null distribution of the homogeneity test exact (multinomial), with no appeal to χ^2 asymptotics. We obtain $\Lambda = 7.13$ on 14 degrees of freedom, exact $p = 0.935$. But its power is such that it reliably resolves only a spread of κ_b with a coefficient of variation of order 0.37 or more (§5.3). The correct conclusion is "base-dependent frequencies are not required by the data", not "they do not exist".

7. **The second-order term of the heuristic itself.** At finite n the LPW heuristic predicts not a constant intensity but $\lambda_b(t) = (\kappa/\ln b)(1 + c_b/t)$: the divisors of $R_b(p)$ are at least $2p + 1$. The exact conditional inference carries over to this model in closed form (§9), but

the exposure must be summed over the prime indices rather than integrated with density $1/\ln n$: for origins $n_{\min} = 2$–19 the integral overstates it by 1.04 in t per base on average (the constant E of Mertens' first theorem) and understates the estimate by 9.0% on the full sample. The positions of the finds determine c_b only weakly, so we measure it directly from the divisibility of $R_b(p)$: $c_b = 0.43$–1.11, with mean 0.67, stable with respect to the threshold; Wagstaff's device gives values twice as large. These values are predicted from first principles—from the primality of the numbers $2kp+1$ and the quadratic character of b (correlation 0.976)—so that c_b depends only on the squarefree part of b. The method has been validated on data generated by the heuristic itself at the level of prime indices: it is unbiased and attains nominal coverage, whereas the constant-intensity model overstates κ by 3–8%. With the calibrated c_b the estimate of κ drops by 3.2% on the full sample and by 5.0–5.8% under truncation, that is, comparably to the cost of the choice of observation scheme. Under (A1), at every lower truncation level one obtains $\hat{\kappa} \approx 1.66$–$1.78$, LPW is not rejected ($p \geq 0.598$), and the same term explains the folklore "slow convergence from below".

8. **An explicit power calculation.** With a two-sided test the pooled sample resolves an excess over e^γ of at least 21.3% and a deficit of at least 19.2%; the Mersenne sequence alone, 45.5% and 37.4% respectively (§6). The test convention is stated explicitly, since the one-sided variant gives a noticeably different threshold (18.7%).

Section 7 works through a narrower but instructive episode: a naive regression of the Mersenne counting function produces an apparently significant negative intercept, which turns out to be an artifact of conditioning. We include it as a worked example of the same error in a different guise.

2 Model and data

2.1 The Poisson model

Put $t = \ln n$. To leading order the heuristic (2) leads to a model in which the indices of generalized repunit primes to base b form a homogeneous Poisson point process on the t-axis with constant intensity

$$\lambda_b = \frac{\kappa}{\ln b}, \qquad \text{LPW: } \kappa = e^\gamma = 1.781072\ldots \tag{3}$$

Universality is the assertion that a single parameter κ governs all bases. The form $\kappa/\ln b$ with a common κ follows from the structure of the divisors of $R_b(p)$. Let p be an odd prime with $p \nmid b-1$. If a prime $q \nmid b-1$ divides $R_b(p)$, then b has order p modulo q, whence $q \equiv 1 \pmod p$, and since $R_b(p)$ is odd, $q \equiv 1 \pmod{2p}$; and if $q \mid b-1$, then $R_b(p) \equiv p \not\equiv 0 \pmod q$. Hence no prime $q < 2p+1$ divides $R_b(p)$ for any such b: $R_b(p)$ has no base-dependent local factors at leading order, of the kind of a Bateman–Horn singular series, and the gain $e^\gamma \ln(2p)$ from the absence of small divisors is the same for all bases (§9). The base enters the leading term only through the size of the number, that is, through $1/\ln b$; dependence on b appears only at second order, through the symbol (b/q) for $q = 2kp+1$, and is accounted for in §9. The Poisson idealization rests on the Cramér model of independent trials and on pairwise coprimality: $\gcd\big(R_b(p), R_b(p')\big) = R_b\big(\gcd(p,p')\big) = 1$ for distinct primes p, p'. It is a working assumption, not a theorem: even for the ordinary primes the Poisson law in short intervals has been derived only conditionally, from the Hardy–Littlewood conjectures [11]; on scales $(\ln x)^A$, $A > 1$, the Cramér model is known to fail [12]; and refinements of the Poisson picture in short intervals remain conjectural [13]. At finite n the heuristic itself specifies not a homogeneous Poisson process but Bernoulli trials at the prime indices with a slowly varying probability; this is accounted for in §9. We adopt (3) as a model and treat κ throughout as a quantity to be estimated; this reduces

the question "is LPW true?" to a one-parameter one and makes the bases poolable. The word "exact" in this paper refers to inference in model (3) and its inhomogeneous generalization: in the Bernoulli-trial model the conditional law of the number of events is Poisson-binomial rather than Poisson, and the Poisson intervals are conservative or practically exact (§9, Remark 4).

2.2 Admissible bases

Bases that are perfect powers are excluded: for $b = c^m$ with $m > 1$ and a prime $n \nmid m$ the number $R_c(n) > 1$ divides $R_b(n)$, so there are only finitely many primes $R_b(n)$ (the index n must divide m), and there is no process of finds at all. Bases $4, 8, 9, 16$ are therefore omitted. For prime n the bases $b \le 26$ have no other infinite algebraic families: for prime n we have $R_b(n) = \Phi_n(b)$, and of the known algebraic factorizations of a cyclotomic polynomial at a b that is not a perfect power only the Aurifeuillean one, at $n = s$, applies (see below). The arithmetic of small indices has a few further finite effects. We note separately one *infinite arithmetic* family of forced composites: if $q = 2p + 1$ is prime, $q \nmid b - 1$ (which certainly holds when $2p + 1 \ge b$) and $(b/q) = +1$, then $q \mid R_b(p)$ (the generalization of the Euler–Lagrange theorem for M_p with $p \equiv 3 \pmod 4$), and $R_b(p)$ is composite except when $R_b(p) = q$ ($b = 2$, $p = 3$). It acts at every p with density of order $1/\ln p$ and is not excluded but enters the second-order term (§9, the case $k = 1$ in (27)). If $p \mid b - 1$, then $R_b(p) \equiv p \pmod{b-1}$ is divisible by p and hence composite. If $p = s$, where $s \equiv 1 \pmod 4$ is the squarefree part of b, the Aurifeuillean factorization applies ($b = 5, 13, 17, 20$). Finally, $R_b(2) = b + 1$ is prime exactly for $b = 2, 6, 10, 12, 18$, so for these five bases the origin $n_{\min} = 2$ is fixed by arithmetic, not by chance. In the main set all these finite effects concern only $n \le 19$; their influence is examined in §5.4 and accounted for in §9. Within the declared range the selection does not depend on the results: we take *all* bases $2 \le b \le 20$ that are not perfect powers, which gives $B = 15$; no base was excluded or added on the strength of its estimate. Each of them has at least eight known indices (the minimum is attained at $b = 18$: eight terms of A133857).

We stress, however, that the upper limit $b \le 20$ is itself a conventional round number and not a consequence of the criterion "at least eight indices": that criterion continues to hold beyond it (A127996 for $b = 21$ has eight terms, A127997 for $b = 22$ has nine) and first fails at $b = 23$ (A204940, six terms). To make sure the limit does not look chosen after the fact, we tested its influence directly. Extending the set to all bases $b \le 26$ that are not perfect powers (this adds $b = 21, 22, 23, 24, 26$; $b = 25 = 5^2$ is excluded algebraically; for $b = 22$ the origin $n_{\min} = 2$ is also fixed by arithmetic, $R_{22}(2) = 23$, and for $b = 24$ the condition $p \mid b-1$ makes $R_{24}(23)$ composite, that is, it concerns the index $23 > 19$) gives $B = 20$, $M = 253$ and

$$\frac{M-1}{S} = 1.9776, \qquad \frac{M-B}{S} = 1.8285,$$

that is, it shifts the main estimate (20) by 0.5%—more than an order of magnitude less than the price of the choice of observation scheme (§4). The homogeneity test on the extended set does not reject homogeneity ($\Lambda = 10.24$ on 19 degrees of freedom, exact $p = 0.953$; tag [S17]). No conclusion of this work depends on the limit $b \le 20$; we retain it as pre-declared.

Remark 1 (Selection on the number of finds). Formally the limit $b \le 20$ was declared in advance, but it was chosen in a world where the numbers of known terms had already been published, and the size of the threat should be stated. The conditional inference of §4 conditions on $N_b \ge 1$—which is harmless: under scheme B with the frontier $W_b = N_b/\lambda_b$ the bias of $(M - B)/S$ is $+0.04\%$. Were a base admitted only with $N_b \ge 7$, the estimate would be biased *upward* by 3.1% (tag [S30]), that is, by an amount of the same order as the residual excess of 3.2%. What protects the conclusion from such selection is not the sign of any shift but the fact that the rule $b \le 20$ does not depend on the number of finds, together with a direct check: lifting the limit to $b \le 26$ adds, in particular, the base $b = 23$ with six terms and shifts the estimate *downward*, from 1.8381

to 1.8285—in the direction predicted by selection on the number of finds, and by an order of magnitude less than the price of the observation scheme.

2.3 The data and their status

Table 1 lists the sequences used, retrieved on 27 August 2026 (cross-check: Appendix A). Four caveats matter.

Probable primes. For large indices most of the entries are probable primes established by a Fermat or strong Fermat test after trial division. PRP status begins far below one million: according to the OEIS entries, from 18 743 for $b = 13$ (A016054), from 19 697 for $b = 14$ (A006032), after 57 917 for $b = 3$ (A028491), from 201 359 for $b = 5$ (A004061). The count is biased upward only if a Fermat pseudoprime has been mistaken for a prime, which for numbers of this size is practically unheard of; the opposite risk, a missed find (below), is more real. Sensitivity to both is examined in §5.4.

Verification frontiers. For $b = 2$ the GIMPS project [14] publishes two frontiers: all exponents below $X_{\text{ver}} = 81\,648\,221$ have been tested twice, which gives exactly 50 Mersenne primes, whereas all exponents below $X_{\text{ft}} = 141\,308\,443$ have been tested at least once, which gives 52. The two largest exponents are provisional: unverified candidates remain between them. The analysis of §7, where the frontier is used explicitly, is restricted to the 50 verified exponents. The pooled analysis is insensitive to this choice (§5.4).

Both frontiers are moving quantities, so we fix the moment of reading: the PrimeNet Milestones report, 27 August 2026. No conclusion depends on the exact value of X_{ver}: it suffices that it lies between $M_{50} = 77\,232\,917$ and $M_{51} = 82\,589\,933$, and this holds for every value the report has taken from the start of 2026 to 4 September 2026. Once double-checking below M_{51} was complete the frontier moved past it (83 200 483 on 30 September 2026); the slice of double-checked exponents then includes M_{51} as well, but the computations of §7 use the snapshot of 27 August 2026, and the pooled analysis does not depend on this slice.

The status of entry A000043. On the retrieval date (27 August 2026) the DATA section of that entry contained 50 terms and stopped at 77 232 917: OEIS regards the order of the subsequent exponents as unproven until double-checking is complete. The exponents 82 589 933 and 136 279 841 are known and documented by GIMPS (and are mentioned in the comments on A000043), but formally they were not part of DATA. After double-checking below 82 589 933 was completed (GIMPS report of 4 September 2026), that exponent was added to DATA as the 51st term (checked on 24 September 2026); this does not affect the computations, since it is already included. We include both exponents, because the statistic (13) uses only the position of the last event and not its ordinal number; §5.4 verifies that restricting base 2 to the fifty double-checked exponents does not change the conclusion.

Search bounds and completeness of coverage for $b > 2$. For the other bases a dated and attributed frontier above the last find has been published only for $b = 10$: "Search limit is 10800000, currently" (S. Batalov, comment on A004023, 1 July 2021). In the extended set it is joined by $b = 22$ and $b = 26$, with the bound 10^5 (R. Price, comments on A127997 and A127999, 2012). For the remaining bases the OEIS entries contain only the dates of the finds, and participants in the distributed search state explicitly that the search limits for most bases are unknown. It is precisely this gap that is the subject of §4 and, we believe, the principal obstacle to a quantitative test of LPW; the four documented frontiers are used in §4.5. We note separately that the entire construction assumes that the search covered $(t_0, \ln L_b]$ *contiguously*. If the sieved ranges contain gaps—a common occurrence in distributed searching—then M is understated

Table 1. Sequences used. N_b is the number of events strictly to the right of $t_0 = \ln n_{\min}$, that is, one less than the number of known prime indices: the smallest index sets the origin, not an event; $K_b = N_b - 1$ is the conditional event count entering (13). The D_b column is rounded, but all computations use exact $D_b = \ln n_{\max} - \ln n_{\min}$. In the last row $M = \sum_b N_b$, $K = M - B$ and $S = \sum_b D_b / \ln b$; the quantity S is not the sum of the D_b column. Source: OEIS entries, retrieved on 27 August 2026; for $b = 2$ the exponents 82 589 933 and 136 279 841 are taken from GIMPS (§2).

b	OEIS	primes	N_b	K_b	$n_{\min}$	$n_{\max}$	D_b	$D_b/\ln b$
2	A000043	52	51	50	2	136 279 841	18.0371	26.0220
3	A028491	23	22	21	3	8 530 117	14.8605	13.5266
5	A004061	19	18	17	3	3 300 593	13.9110	8.6434
6	A004062	17	16	15	2	3 360 347	14.3344	8.0002
7	A004063	10	9	8	5	1 264 699	12.4409	6.3934
10	A004023	11	10	9	2	8 177 207	15.2237	6.6116
11	A005808	13	12	11	17	1 868 983	11.6077	4.8408
12	A004064	14	13	12	2	769 543	12.8604	5.1754
13	A016054	13	12	11	5	1 503 503	12.6139	4.9178
14	A006032	11	10	9	3	1 724 417	13.2618	5.0252
15	A006033	10	9	8	3	639 833	12.2704	4.5311
17	A006034	12	11	10	3	1 990 523	13.4053	4.7315
18	A133857	8	7	6	2	1 270 141	13.3615	4.6228
19	A006035	11	10	9	19	209 359	9.3074	3.1610
20	A127995	9	8	7	3	984 349	12.7011	4.2397
total			$M = 218$	$K = 203$				$S = 110.4423$

relative to the exposure and $\hat{\kappa}$ is biased downward, that is, in the same direction as our main conclusion. In general, completeness of coverage cannot be checked from published data; how large such a gap would have to be to change the conclusion is assessed in §5.4. That such gaps are real is shown by base 5. In 2024 the probable primes $R_5(4\,939\,471)$ and $R_5(5\,154\,509)$ were found (R. Propper); they are not in the OEIS, and summary tables mark the range 3 300 593–4 939 471 as not searched contiguously. For the conditional inference this matters: law (12) requires contiguous coverage of $(t_0, t_0 + D_b)$. We therefore retain $n_{\max} = 3\,300\,593$. Including both numbers, which is admissible only under contiguous coverage, would give $\hat{\kappa} = 1.852$ instead of 1.838 (tag [S18]). This is one more argument for publishing not only the frontiers but also the sieving protocols.

3 What is biased and what is not

3.1 The estimators in common use

Dubner [4] and Bourdelais [5] characterize a base b by a normalized occurrence frequency obtained by fitting a straight line to $\ln n_k$ against k and dividing by $\ln b$. By (3) the expected slope is $1/\lambda_b = \ln b/\kappa$, so the normalized frequency estimates

$$G_b = \frac{1}{\kappa}, \qquad \text{LPW:}\ \ G_b = e^{-\gamma} = 0.5614595\ldots \tag{4}$$

Several variants are in circulation—ordinary OLS with an intercept, regression through the origin, and the endpoint form $\hat{G}_b = D_b/(N_b \ln b)$ with $D_b = t_{N_b} - t_0$. All of them share one structural feature: the fitting interval ends at the largest known index. Regression through the origin is understood in two ways. The line through the point $(0, t_0)$, that is, OLS of $t_k - t_0$ on k without an intercept, has weights $w_k = k/\sum_j j^2$ with $\sum_k w_k k = 1$ and is unbiased under scheme A by

the same argument as Proposition 2. The line through the literal origin of the $(k, \ln n_k)$ plane is biased: $\mathbb{E}[\hat{\beta}] = 1/\lambda + 3t_0/(2N+1)$, which is from 5.2% ($b = 2$) to 25.4% ($b = 19$) of $1/\lambda$ (tag [S30]); this variant is not considered below.

3.2 The exact distribution

Proposition 1 (Conditional law of the right endpoint)**.** *Let the events form a homogeneous Poisson process of intensity λ on (t_0, ∞), and let $t_1 < \cdots < t_N$ be its first N events after t_0. Then $D := t_N - t_0$ has the* Gamma(N, λ) *distribution with density $\lambda^N d^{N-1} e^{-\lambda d}/(N-1)!$. Conditionally on $D = d$, the preceding $N-1$ events are distributed as the order statistics of $N-1$ independent uniform variables on $(t_0, t_0 + d)$.*

Proof. The waiting times $t_1 - t_0, \ldots, t_N - t_{N-1}$ are independent Exp(λ) variables, and a sum of N of them is Gamma(N, λ). The joint density of $(t_1, \ldots, t_N)$ on $t_0 < t_1 < \cdots < t_N$ is the product of the densities of the spacings, that is, $\lambda^N e^{-\lambda(t_N - t_0)}$. Dividing by the Gamma$(N, \lambda)$ density at $d = t_N - t_0$ gives the conditional density $(N-1)!/d^{N-1}$ of the first $N-1$ events on $t_0 < t_1 < \cdots < t_{N-1} < t_0 + d$, which is the density of the order statistics of $N-1$ independent uniform variables. □

In the data $t_0 = \ln n_{\min}$ is the position of the first find for ten of the fifteen bases, and is fixed by arithmetic for five ($b = 2, 6, 10, 12, 18$, §2.2). This does no harm: the time of the first event is a stopping time, and by the strong Markov property the process to its right is again Poisson with the same intensity. The same applies to Propositions 2–4 and to (14).

Corollary 1 (Bias of the reciprocal)**.** *The maximum-likelihood estimator of λ based on D alone is $\hat{\lambda} = N/D$, and*

$$\mathbb{E}[\hat{\lambda}] = N\,\mathbb{E}[D^{-1}] = \frac{N}{N-1}\,\lambda, \tag{5}$$

since $\mathbb{E}[D^{-1}] = \lambda/(N-1)$ for $D \sim$ Gamma(N, λ) with $N \geq 2$. The estimator $\tilde{\lambda} = (N-1)/D$ is exactly unbiased.

3.3 What does not follow from Corollary 1

It is natural to conclude that, since $\hat{\lambda}$ is inflated by the factor $N/(N-1)$, the frequency $\hat{G}_b$ must be deflated by the same factor. This is false under either of the two observation schemes analyzed below.

Proposition 2 (Unbiasedness of the frequency)**.** (a) Scheme A. *Under the hypotheses of Proposition 1:*

(i) *the endpoint estimator is exactly unbiased:* $\mathbb{E}\left[\dfrac{D}{N \ln b}\right] = \dfrac{1}{\lambda \ln b} = G_b$;

(ii) *the OLS slope in the regression of t_k on k is exactly unbiased:* $\mathbb{E}[\hat{\beta}] = 1/\lambda$;

(iii) *consequently the "corrected" quantity $\tilde{G}_b = \frac{N}{N-1}\hat{G}_b$ is biased* upward*, by exactly the factor $N/(N-1)$.*

(b) Scheme B. *Let the process be observed on a window of width W, and put $\mu = \lambda W$; the estimators are set to zero where they are undefined. Then*

$$\mathbb{E}\left[\frac{D}{N}\,\mathbf{1}_{\{N \geq 1\}}\right] = \mathbb{E}\left[\hat{\beta}_0\,\mathbf{1}_{\{N \geq 1\}}\right] = \frac{1}{\lambda}\Big(1 - (1+\mu)e^{-\mu}\Big), \tag{6}$$

where $\hat\beta_0$ is the OLS slope through the points (k, t_k), $k = 0, \dots, N$, including *the origin. For the slope $\hat\beta_1$ through the points $k = 1, \dots, N$* without *the origin, which is defined for $N \geq 2$,*

$$\mathbb{E}\big[\hat\beta_1 \mid N \geq 2\big] = \frac{1}{\lambda} \cdot \frac{1 - \big(1 + \mu + \mu^2/2\big)e^{-\mu}}{1 - (1+\mu)e^{-\mu}}. \tag{7}$$

Both forms are thus unbiased up to a term exponentially small in μ; at $\mu = 7$ the deficit is 0.73% *in* (6) *and* 2.25% *in* (7)*, and at $\mu = 10$ it is* 0.05% *and* 0.23%*.*

Proof. (a)(i) $\mathbb{E}[D] = N/\lambda$ for $D \sim \text{Gamma}(N, \lambda)$, whence the result at once. (ii) The OLS slope is the linear statistic $\hat\beta = \sum_k w_k t_k$ with weights $w_k = (k - \bar k)/\sum_j (j - \bar k)^2$, for which $\sum_k w_k = 0$ and $\sum_k w_k k = 1$. Since $\mathbb{E}[t_k] = t_0 + k/\lambda$, we get $\mathbb{E}[\hat\beta] = \sum_k w_k\,(t_0 + k/\lambda) = 1/\lambda$. (iii) is immediate from (i).

(b) Under scheme B, conditionally on $N = n \geq 1$, the positions of the events are the order statistics of n independent uniform variables on $(t_0, t_0 + W]$, so $D/W \sim \text{Beta}(n, 1)$ and $\mathbb{E}[D \mid N = n] = Wn/(n+1)$, whence $\mathbb{E}[D/N \mid N = n] = W/(n+1)$. Averaging over $N \sim \text{Pois}(\mu)$ and taking into account that D is defined only for $N \geq 1$,

$$\mathbb{E}\Big[\frac{D}{N}\Big] = W \sum_{n \geq 1} \frac{\mu^n e^{-\mu}}{n!\,(n+1)} = \frac{W}{\mu} \sum_{n \geq 1} \frac{\mu^{n+1} e^{-\mu}}{(n+1)!} = \frac{W}{\mu}\Big(1 - e^{-\mu} - \mu e^{-\mu}\Big),$$

which is (6) for the endpoint form. For the OLS slope, note that $\mathbb{E}[t_k \mid N = n] = t_0 + kW/(n+1)$ for $k = 0, \dots, n$; since the slope is a linear statistic with weights $\sum_k w_k = 0$, $\sum_k w_k k = 1$, in both variants $\mathbb{E}[\hat\beta \mid N = n] = W/(n+1)$. For $\hat\beta_0$ the summation runs over $n \geq 1$ and gives the same result as above. For $\hat\beta_1$ it runs over $n \geq 2$: $\sum_{n \geq 2} \frac{W}{n+1} \frac{\mu^n e^{-\mu}}{n!} = \frac{W}{\mu}\big(1 - (1 + \mu + \mu^2/2)e^{-\mu}\big)$, and division by $\Pr(N \geq 2) = 1 - (1+\mu)e^{-\mu}$ gives (7). □

The substantive point is simple: unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another, because $x \mapsto 1/x$ is strictly convex. The unbiased pair is $(\hat G_b,\ \tilde\kappa_b)$ with $\tilde\kappa_b = (N_b - 1)\ln b / D_b$; but $\tilde\kappa_b \neq 1/\hat G_b$. Any single "censoring correction" that repairs one of these quantities damages the other. Table 2 confirms Proposition 2 by simulation under both schemes.

Remark 2. Part (b) matters because it removes a possible objection to the first conclusion of this work. Below (§4) we adopt scheme B as matching the way the data were obtained; if part (a) were the only one, the whole of §3 would rest on a scheme that the paper itself rejects. Part (b) shows that the conclusion does not depend on the choice of scheme: the deficit $(1+\mu)e^{-\mu}$ at $\mu \approx N_b$ is 0.73% for $N_b = 7$, 0.05% for $N_b = 10$ and about $2 \cdot 10^{-6}$ for $N_b = 16$ (for OLS without the origin, (7): 2.25%, 0.23% and $1.4 \cdot 10^{-5}$), whereas the deviations discussed in the literature are tens of percent.

Remark 3. The negative result of this subsection deserves emphasis, because it runs against intuition: the systematic downward deviation of the published frequencies from $e^{-\gamma}$ is *not* explained by the fitting scheme. Its source is not the arithmetic of the estimator but the second-order term of the heuristic itself, $c_b \ln\ln x$ in the counting function, which depresses the frequency at finite n and vanishes only as $\ln\ln n / \ln n$ (§9); the observation scheme also contributes to the gap (§4).

3.4 Instability with respect to the form of the estimator

Unbiasedness does not mean agreement. Table 5 gives the endpoint form $\hat G_b$ and the OLS estimate $\hat G_b^{\text{OLS}}$ on identical data. The discrepancies reach 29.5% ($b = 18$), 15.9% ($b = 13$) and 10.3% ($b = 11$).

Compare these with what the literature interprets substantively—the deviations of the estimates themselves from $e^{-\gamma}$. Across the fifteen bases the latter have median 17.6% (from 5.6% at $b = 20$ to 43.7% at $b = 19$). The largest discrepancy between forms, 29.5%, exceeds that

Table 2. Monte Carlo check of Proposition 2, 60 000 replications per row (seed fixed, see `analysis_v2.py`, tag [S1]); the Monte Carlo standard error is about 0.002 at $N = 7$ and 0.0006 at $N = 51$ (tag [S29]), so the fourth digit is noise. Target values: $N/(N-1)$ for $\mathbb{E}[\hat\lambda]/\lambda$; the column "theory" and (7) for the scheme B columns; 1.0000 for the rest. In the right-hand part $\mu = \lambda W$; the column "theory" gives the factor $1-(1+\mu)e^{-\mu}$ of (6) for $\hat G$. The scheme B column $\mathbb{E}[\hat\beta]\lambda$ is for the slope $\hat\beta_1$ without the origin, conditional on $N \ge 2$; its theoretical value is given by (7): 0.9775, 0.9977 and 1.0000 at $\mu = 7, 10, 16$.

	scheme A (N fixed)					scheme B ($\mu = \lambda W$)		
$N = \mu$	$\mathbb{E}[\hat\lambda]/\lambda$	$N/(N-1)$	$\mathbb{E}[\tilde\lambda]/\lambda$	$\mathbb{E}[\hat G]\lambda$	$\mathbb{E}[\hat\beta]\lambda$	$\mathbb{E}[\hat G]\lambda$	theory	$\mathbb{E}[\hat\beta]\lambda$
7	1.1655	1.1667	0.9990	1.0012	1.0020	0.9908	0.9927	0.9790
10	1.1103	1.1111	0.9992	1.0007	1.0018	0.9997	0.9995	0.9948
16	1.0650	1.0667	0.9985	1.0016	1.0017	0.9992	1.0000	0.9992
22	1.0462	1.0476	0.9987	1.0011	1.0006	0.9991	1.0000	0.9992
51	1.0206	1.0200	1.0006	0.9994	0.9992	0.9997	1.0000	0.9999

median by a factor of 1.7. On average the choice of form is of course cheaper: the median discrepancy is 4.1%, about a quarter of the median deviation. But for two of the fifteen bases ($b = 18$ and $b = 20$) the choice of form weighs more than their entire deviation from the LPW prediction—and one of them, base 18, is precisely the one the folklore discusses.

The consequence for the folklore is direct. Under the endpoint form base 18 has $\hat G_{18} = 0.6604$, the third largest of fifteen (after $b = 7$ with 0.7104 and $b = 10$ with 0.6612), so it belongs firmly among the three "unluckiest"; under OLS, $\hat G_{18}^{\mathrm{OLS}} = 0.4656$, tenth of fifteen and below the median 0.4904, so it has become "lucky". The label flips without a single new prime, purely from a change in the form of the estimator: the base moves from third place to tenth.

We must add, however, that so large a discrepancy for base 18 is created by a single feature of the data. The origin $n_{\min} = 2$ is fixed here by arithmetic ($R_{18}(2) = 19$, §2.2), and the next index, 25 667, lies 70.8% of the whole window in t away from it. For conditionally uniform interior points the probability that the largest of seven gaps is at least this long is 0.0043 (about 6% after a Bonferroni correction for fifteen bases). Measured from 25 667, the two forms give 0.2250 and 0.2135, a discrepancy of only −5.1% (tag [S29]). The general point—the label describes the choice of estimator—stands, but its quantitative illustration rests on one anomalous gap.

To this is added a second, technical difficulty with OLS: its standard errors are wrong in this problem. The residuals are deviations of order statistics and are strongly autocorrelated; for $b = 2$ we obtain $\rho_1 = 0.825$, and the Newey–West covariance estimator with eight lags raises the standard error of the intercept from 0.447 to 0.732 (§7). Published point values come with no intervals at all, and this—not bias—is the principal methodological objection to current practice.

3.5 The conditional likelihood under scheme A

Proposition 1 yields more than an analysis of bias. Setting $t_0 = \ln n_{\min}$ and taking (N_b, D_b) as the data for base b, we obtain the log-likelihood

$$\ell_{\mathrm{A}}(\kappa) = \sum_b \Big[N_b \ln \frac{\kappa}{\ln b} - \frac{\kappa}{\ln b} D_b \Big] + \text{const}, \tag{8}$$

whose distribution does not depend on the search bound L_b. This matters in practice: in the main set L_b is documented only for $b = 2$ and $b = 10$. Differentiating (8) and putting $S = \sum_b D_b / \ln b$, we get

$$\hat\kappa = \frac{M}{S}, \qquad \tilde\kappa = \frac{M-1}{S}, \qquad M = \sum_b N_b. \tag{9}$$

Since $\kappa S = \sum_b \lambda_b D_b$ is a sum of independent Gamma$(N_b, 1)$ variables,

$$\kappa S \sim \text{Gamma}(M, 1), \tag{10}$$

an exact pivot; confidence intervals and p-values follow in closed form. *It is precisely this statement that requires revision*—see §4.

Remark 4 (Independence across bases). Pooling presupposes that the processes for different bases are independent. Strictly this fails at small indices: seventeen index values occur in more than one base, and $n = 3$ occurs in nine bases, since $R_b(3) = b^2 + b + 1$ is prime for many b. All such coincidences lie at $n \leq 317$. More important than coinciding indices (the numbers $R_b(n)$ for different b are distinct) is the possible sharing of divisors. It affects neither only small n nor only the upper end of the range: $D_b = t_{N_b} - t_0$ depends also on the origin, which for eleven of the fifteen bases lies at $n_{\min} \in \{2, 3\}$, that is, precisely in the zone of coincidences. There is also a structural dependence at every n: for a given p the candidate divisors $2kp + 1$ are common to all bases, and bases with the same squarefree part (the pairs 2 and 18, 3 and 12, 5 and 20) also share the quadratic character (b/q), hence the same second-order term (§9). This covariance can be estimated within the same heuristic. Take a crude model in which the primality of $q = 2p+1$ is a random variable common to all bases: if q is prime and $(b/q) = +1$, then $R_b(p)$ is composite, otherwise a Bernoulli trial takes place. Then for $p > 10$ the covariance of the event counts is 0.146 for the pair 2 and 18, 0.072 for 3 and 12 and 0.068 for 5 and 20, and all 105 pairs together increase $\text{Var}(K)$ by 3.0%. This is offset by the opposite effect: in the Bernoulli-trial model the variance is below the Poisson one. In the same model (primes $p > 10$ in the windows, without conditioning on D_b) the ratio $\text{Var}(K)/\mathbb{E}[K]$ is 0.961 without the covariance and 0.990 with it (tag [S30]), so the Poisson law is practically exact. At small n the issue is not stochastic dependence at all but deterministic arithmetic: the primality of $b+1$ and b^2+b+1 is given, not random, and the right response is to move the origin above that zone. Subsection 5.4 shows that discarding all indices below 10^3, moving the origin, and excluding any single base leave the conclusion unchanged. We note that this remark bears more strongly on the conditional multinomial test of §5.3 than on the pooled estimate: dependence distorts not the mean but the null distribution of the statistic; the same truncation check covers that test as well.

4 Two observation schemes

4.1 The distinction

The pivot (10) was derived on the assumption that N_b is fixed by the observation plan: we watch the process until N_b events have accumulated, then stop. Call this *scheme A* (inverse sampling, Type-II censoring).

The data arose differently. For each base the computational search covered the whole interval $(t_0, \ln L_b]$, and every event falling inside it was found; the number of finds N_b is a random variable, while the frontier L_b is fixed. Call this *scheme B*.

Remark 5 (What here is classical and what is not). The distinction between the schemes is itself classical and goes back to reliability theory, where scheme A is called a *failure-truncated* and scheme B a *time-truncated* life test [15, 16]. The estimator $(N-1)/D$, unbiased under scheme A (Corollary 1), is a textbook fact there, and we lay no claim to it. The distribution of the MLE of the mean under Type-I censoring was studied by Bartholomew [17]. We also note an elementary fact: under scheme B with a known window the sufficient statistic is $N \sim \text{Poisson}(\lambda W)$, and $\mathbb{E}f(N) = \sum_n f(n)e^{-\mu}\mu^n/n!$ is an entire function of $\mu = \lambda W$, so it cannot coincide with W/μ, which has a pole at zero. Hence no unbiased estimator of the *mean* $1/\lambda$ exists under scheme B (for time-censored samples from the exponential distribution see [18]); the same argument applies conditionally on D_b. Proposition 3 does not contradict this: it gives exact unbiasedness of an estimator of the *intensity*.

The mathematical apparatus is classical as well: law (12) and the loss of one event per stratum follow from the general theory of conditional likelihood with nuisance parameters (Remark 7). What is new lies in the setting and the application. In the classical setting the moment at which the test is stopped is *known*: the time T is set by the experimenter. Here, in the main set, the frontier L_b is documented only for $b = 2$ and $b = 10$ (§4.5), so the full exposure cannot be read off a protocol and must instead be reconstructed—via the memorylessness of the process—from the finds themselves. Hence the correction is needed *per base*, rather than for the sample as a whole: there are as many unknown frontiers as there are bases. A survey of the biases generated by the edge of the observation window, though in a nonparametric setting and for the shape of the interevent distribution rather than for the intensity, is given in [19].

Dual problems have long been known. In paleontology the unknown end of a stratigraphic range is estimated from fossil finds, and the observed extreme horizon is extended by roughly one mean gap between finds [20, 21]. In ecology the unknown extinction time of a species is estimated from its sighting record [22]. The general problem of estimating a truncation point is treated in [23]. There, however, the unknown end of the window is the parameter of interest. Here it is a nuisance parameter, and it is the intensity that is estimated. Closest in technique is [24]: there the observation window starts at the first event, and the likelihood factorizes into a Poisson factor and a factor that does not depend on the intensity. We have found no work in which Type-I censoring with an unrecorded censoring time is treated as a problem of pooled estimation of an intensity across many strata.

The distinction is not cosmetic. Under scheme B, conditionally on $N_b = n$, the positions of the events are n independent uniform variables on $(t_0, \ln L_b]$, so the conditional distribution of D_b depends on L_b and *not* on λ_b: with N_b fixed, D_b is informative about the frontier, not about the intensity.

More precisely, under scheme B with a known window width $W_b = \ln L_b - t_0$ the likelihood for base b is $\lambda_b^{N_b} e^{-\lambda_b W_b}$, whereas with an unknown width it depends on the nuisance parameter $W_b \geq D_b$ and decreases monotonically in it. The joint MLE therefore sets $\hat{W}_b = D_b$—the frontier exactly at the last find—and expression (8) turns out to be the *profile* likelihood over the nuisance parameters $W_1, \ldots, W_B$, with M/S the profile MLE. The number of nuisance parameters grows with the number of bases, and the profile MLE is inconsistent: as $B \to \infty$ with $\lambda_b W_b$ bounded, it does not converge to κ. This is the classical Neyman–Scott situation [9] (for a survey see [25]), and the standard way out is to condition on a statistic sufficient for the nuisance parameters [26, 10]. This is what §4.2 implements.

Note also that §7 of the present work, where the frontier for $b = 2$ is known and used explicitly, rests on scheme B. One cannot apply both schemes in a single paper; below we adopt scheme B as matching the way the data were obtained, and construct exact inference for it.

4.2 The exact conditional law under scheme B

The key observation is that under scheme B one should condition not on the number of events but on the position of the last of them. The unknown frontier then disappears from the problem entirely. One assumption is required.

(A1) *The position of the frontier L_b was not chosen in the light of the search results: the computational budget, not the finds, determines how far in n the sieving has been carried. Formally: the window width $W_b = \ln L_b - t_0$ is a random variable with an arbitrary unknown distribution π_b (in particular, a constant), and the whole vector $(W_b)_b$ is independent of the totality of the processes of finds for all bases. For law (12) for a single base it suffices that W_b be independent of the process for base b; joint independence is needed for part (iii) and for the multinomial test of §5.3.*

In the language of survival analysis, (A1) is the requirement that the censoring be *non-informative*: the stopping time does not depend on what the process has produced. The

distributions π_b may be arbitrary and may differ between bases; they are never estimated.

Proposition 3 (Exact conditional likelihood)**.** *Suppose that under scheme B a homogeneous Poisson process of intensity λ_b is observed on $(t_0, t_0 + W_b]$, that (A1) holds and $N_b \geq 1$, and let $D_b = t_{N_b} - t_0$ be the distance from the origin to the last event found.*

(i) *The joint density of (N_b, D_b) at (n, d), $n \geq 1$, $d > 0$, factorizes:*

$$p(n,d) = \underbrace{e^{-\lambda_b d}\frac{(\lambda_b d)^{n-1}}{(n-1)!}}_{\text{kernel of the conditional law}} \cdot \underbrace{\lambda_b \int_{[d,\infty)} e^{-\lambda_b (w-d)}\, \pi_b(dw)}_{\text{(defective) density of } D_b}, \tag{11}$$

so that for fixed λ_b the statistic D_b is sufficient for W_b (and for the whole distribution π_b).

(ii) *Consequently,*

$$K_b := N_b - 1 \;\big|\; D_b \;\sim\; \text{Poisson}(\lambda_b D_b), \tag{12}$$

and this distribution depends neither on W_b nor on π_b.

(iii) *For processes that are independent across bases,*

$$K := M - B \;\big|\; (D_b)_b \;\sim\; \text{Poisson}(\kappa S), \qquad S = \sum_b \frac{D_b}{\ln b}, \tag{13}$$

and $\hat{\kappa}_{\mathrm{B}} = (M-B)/S$ is the conditional maximum-likelihood estimator, exactly conditionally unbiased: $\mathbb{E}[\hat{\kappa}_{\mathrm{B}} \mid (D_b)_b] = \kappa$.

Proof. (i) First fix $W_b = w$. For $0 < d < d + \Delta \leq w$ and $\Delta \to 0$, the event $\{N_b = n,\ D_b \in [d, d+\Delta)\}$ means: exactly $n-1$ events in $(t_0, t_0 + d)$, one in $[t_0 + d, t_0 + d + \Delta)$ and none in $[t_0 + d + \Delta, t_0 + w]$. By the independence of the increments of the Poisson process its probability equals $e^{-\lambda_b d}\frac{(\lambda_b d)^{n-1}}{(n-1)!} \cdot \lambda_b \Delta \cdot e^{-\lambda_b (w-d)} + o(\Delta)$, and for $d > w$ it is zero. By (A1), averaging over $w \sim \pi_b$ affects only the last factor and gives (11). Summation over $n \geq 1$ shows that the second factor in (11) is the marginal density of D_b; it is defective: its total mass is $\Pr(N_b \geq 1) < 1$, since D_b is defined only when there are finds. (ii) Divide (11) by the marginal density of D_b. (iii) A sum of independent Poisson variables is Poisson; the conditional log-likelihood $\ell_{\mathrm{B}}(\kappa) = \sum_b \big[K_b \ln \kappa - \kappa D_b / \ln b\big] + \text{const}$ is maximized at $\kappa = K/S$, and conditional unbiasedness is immediate from $\mathbb{E}[K \mid D] = \kappa S$. □

Remark 6 (Conditional and integrated likelihood). The same result is obtained by integrating the likelihood over the nuisance parameter. For a fixed window the joint density is proportional to $\lambda_b^n d^{n-1} e^{-\lambda_b W_b} \mathbf{1}_{\{W_b \geq d\}}$, and $\int_d^\infty \lambda_b^n e^{-\lambda_b W_b}\, dW_b = \lambda_b^{n-1} e^{-\lambda_b d}$, which is exactly the kernel of (12). This is the integrated likelihood in the sense of [27] with an improper uniform measure on W_b. Such a procedure depends, in itself, on the choice of measure: a measure uniform in $\mu_b = \lambda_b W_b$, that is, $d\mu_b = \lambda_b\, dW_b$, would give $\lambda_b^n e^{-\lambda_b d}$ and bring back the inconsistent estimator M/S. Here, however, the choice is not arbitrary. First, the result coincides with the *conditional* likelihood (12), which is defined without any measure and is invariant under reparametrization of W_b. Second, W_b enters (11) as a location parameter for $D_b = W_b - \delta_b$ (up to the truncation $\delta_b \leq W_b$), and dW_b is the invariant measure of the translation group. Third, for $\pi_b(dw) = dw$ the marginal factor in (11) equals $\lambda_b \int_d^\infty e^{-\lambda_b (w-d)} dw = 1$ and does not depend on λ_b. Strictly speaking, D_b is not S-ancillary: the marginal density $h(d)$ in (11) is subject to a constraint—the function $h(d) e^{-\lambda_b d} = \lambda_b \int_{[d,\infty)} e^{-\lambda_b w}\, \pi_b(dw)$ is non-increasing in d—and this constraint depends on λ_b. Hence the factor discarded by conditioning carries some information about κ, but only through a shape constraint, which with a single value of D_b per base is practically worthless and cannot be used without specifying π_b (compare the optimality conditions for the conditional estimator in [28]).

Remark 7 (Connection with the Neyman–Scott problem). The correction "one event per base" is of the same nature as the loss of one degree of freedom per group in the classical Neyman–Scott example [9]. For $y_{ij} \sim N(\mu_i, \sigma^2)$ the profile MLE $\hat{\sigma}^2$ is inconsistent, whereas conditioning on $\bar{y}_i$, which is sufficient for μ_i at fixed σ^2, gives $\sum_j (y_{ij} - \bar{y}_i)^2 \sim \sigma^2 \chi^2_{n_i - 1}$. In both cases the nuisance parameter of a group is one-dimensional and is a location parameter, and the statistic sufficient for it "absorbs" exactly one observation: the mean absorbs one degree of freedom, the last event absorbs one event. The general mechanism is the conditional likelihood given a statistic that is sufficient for the nuisance parameter at a fixed value of the parameter of interest [26]. Its consistency as the number of strata grows was established under regularity conditions by Andersen [10], and the optimality of the conditional estimator in models that mix over the nuisance parameter by Lindsay [28].

Remark 8 (Inhomogeneous intensity). Argument (i) does not use homogeneity. If the intensity $\lambda_b(t; \theta)$ depends on a parameter θ, then conditionally on D_b the events in $(t_0, t_0 + D_b)$ form a Poisson process of the same intensity, and the conditional log-likelihood is

$$\ell_{\mathrm{B}}(\theta) = \sum_b \Bigl[\sum_{i=1}^{N_b - 1} \ln \lambda_b(t_i; \theta) - \int_{t_0}^{t_0 + D_b} \lambda_b(t; \theta)\, dt \Bigr] : \tag{14}$$

the last event is excluded from the sum, and the integral stops at it. It is in this form that the variable-intensity models of §8 should be estimated.

The practical consequences of Proposition 3 deserve to be listed separately, since they remove almost all the caveats of the previous version of this work.[1]

1. *Exact intervals without simulation.* Inverting the exact Poisson test for the rate gives the Garwood interval [30]
$$\Bigl[\tfrac{1}{2S} \chi^2_{\alpha/2,\, 2K},\ \tfrac{1}{2S} \chi^2_{1-\alpha/2,\, 2(K+1)} \Bigr], \tag{15}$$
whose coverage is never below the nominal level. For our S it equals 0.9540 at $\kappa = e^{\gamma}$ and lies within 0.950–0.958 for $\kappa \in [1.5,\ 2.2]$ (the excess is the usual effect of discreteness).

2. *No remainder.* No exponentially small terms, no caveats about bases without finds: conditioning on $N_b \geq 1$ holds by construction, since D_b is defined only then. A base without a single find does not enter the conditional likelihood.

3. *No dependence on the position of the frontier.* The rule by which L_b was chosen does not enter (12) at all—provided (A1) holds.

4. *The cost of conditioning is small.* $\mathrm{SD}(\hat{\kappa}_{\mathrm{B}}) = \sqrt{\kappa / S} = 0.1270$ against $\sqrt{\kappa / S_{\mathrm{full}}} = 0.1224$ with known frontiers standing at their expected positions, $S_{\mathrm{full}} \approx S + B/\kappa$ (Proposition 4). Forgoing knowledge of L_b increases the standard deviation by 3.7%, or costs 7.1% of the Fisher information: B/κ out of $S + B/\kappa$, that is, roughly B events out of M. Knowledge of the frontiers is needed not so much for precision as for checking (A1) (§4.5).

If (A1) is violated, conditioning on D_b ceases to be harmless. If the search stops exactly at a find, then D_b is itself the stopping time, $\delta_b = 0$, and we are back to scheme A. If instead the search stops after a sufficiently long stretch without finds, then the stopping itself carries information about the process before D_b, and law (12) no longer holds (Remark 10).

Remark 9 (Actual and published frontier). Condition (A1) concerns the actual search horizon T_b—the bound up to which the sieving has really been carried as of the snapshot—and not a number that someone has published. If T_b satisfies (A1) and all finds on $(t_0, \ln T_b]$ have been published

[1] The previous versions and their computational scripts are openly available in the repository history [29]; here and below, references to the "previous" or an "earlier" version are to them.

by the snapshot date, then (N_b, D_b) are determined by the search on that window, and law (12) holds however the published bound L_b, $\ln L_b \in [t_0 + D_b, \ln T_b]$, is chosen: it may be reported soon after a find, rounded, or left unchanged for years. The published gap $\ln L_b - t_{N_b}$ is then only a lower bound on the true δ_b. Condition (A1) itself, on the other hand, cannot be weakened within the class of stopping times to, say, "the stopping density depends on the finds only through the position of the last of them". Let $Q(w) = q(w, d(w))$, where $d(w)$ is the last find before w. Insert an extra find at a point s between neighbouring finds $d_i < s < d_{i+1}$. The normalization $\int Q\, dw = 1$ must hold for both configurations, whence $\int_s^{d_{i+1}} [q(w,s) - q(w,d_i)]\, dw = 0$; varying the position of d_{i+1} gives the same for every upper limit $x > s$, that is, $q(w,s) = q(w,d_i)$ for almost all w: the density does not depend on the finds, which is (A1) again. The rules "stop at the next find" and "stop after a drought" depend on the configuration and violate law (12) (Remark 10).

4.3 What ignorance of the frontier costs

Besides Remark 7, the size of the correction is explained by a direct moment calculation. We retain it as a substantive interpretation and as the basis for the analysis of violations of (A1) in Remark 10.

Proposition 4. *Suppose that under scheme B the search for base b covered $(t_0, \ln L_b]$, that the window width $W_b = \ln L_b - t_0$ is fixed (under (A1), conditionally on W_b), and let $\delta_b = \ln L_b - t_{N_b}$ be the distance from the frontier to the last event found (with $D_b = 0$ and $\delta_b = W_b$ when $N_b = 0$, so that always $D_b + \delta_b = W_b$). Then*

$$\Pr(\delta_b > x) = e^{-\lambda_b x} \quad (0 \le x < W_b), \qquad \Pr(\delta_b = W_b) = e^{-\lambda_b W_b}, \tag{16}$$

that is, δ_b is an exponential variable with parameter λ_b truncated at W_b, and

$$\mathbb{E}[\delta_b] = \frac{1 - e^{-\lambda_b W_b}}{\lambda_b} \;\le\; \frac{1}{\lambda_b}. \tag{17}$$

The full exposure is $S_{\text{full}} = S + \sum_b \delta_b / \ln b = \sum_b W_b / \ln b$, with $\mathbb{E}[M] = \kappa S_{\text{full}}$ identically, and

$$\mathbb{E}\Big[\kappa \sum_b \delta_b / \ln b\Big] = B - \sum_b e^{-\lambda_b W_b}. \tag{18}$$

Proof. For $x < W_b$ the event $\{\delta_b > x\}$ means that the interval $(\ln L_b - x, \ln L_b]$ of length x contains no event of the process, which has probability $e^{-\lambda_b x}$. As $x \to W_b^-$ what remains is the case "no events at all", which gives the atom in (16). Integrating the tail, $\mathbb{E}[\delta_b] = \int_0^{W_b} e^{-\lambda_b x}\, dx = (1 - e^{-\lambda_b W_b})/\lambda_b$, whence (17); the inequality is obvious. Next, S_{full} is not random: by construction $D_b + \delta_b = W_b$, so $\kappa S_{\text{full}} = \sum_b \lambda_b W_b = \mathbb{E}[M]$—this is exactly the moment equation. Finally $\kappa\, \mathbb{E}[\delta_b]/\ln b = \lambda_b \mathbb{E}[\delta_b] = 1 - e^{-\lambda_b W_b}$, and summation over b gives (18). □

In other words, ignorance of the frontier costs information equivalent to one event *per base*—whereas formula (9) subtracts one event *in total*. With $B = 15$ the difference is fourteen events, that is, 6.9%:

$$\frac{M - 1}{S} = 1.9648 \qquad \text{versus} \qquad \frac{M - B}{S} = 1.8381. \tag{19}$$

The remainder $\sum_b e^{-\lambda_b W_b}$ dropped in (18) is the expected number of bases for which the search produced no find at all. The plug-in $\lambda_b W_b \approx N_b \ge 7$ estimates it by $\sum_b e^{-N_b}$, about $2 \cdot 10^{-3}$ against $B = 15$; this is an estimate, not a bound, since the $\lambda_b W_b$ are unknown. We stress that in the exact formulation (12) this remainder does not arise at all: the moment calculation is needed here only for interpretation.

Remark 10 (What remains without (A1)). The previous version claimed that the family $(M - Bm)/S$ with $m \in [0, 1]$ yields a provable two-sided bracket $[(M-B)/S,\ (M-1)/S]$. This claim is *false*, and we withdraw it. Inequality (17) was derived for a fixed window (or one independent of the process), that is, precisely under (A1), and then no bracket is needed: the estimator $(M - B)/S$ is correct. Put $m_b = \lambda_b \mathbb{E}[\delta_b]$: under (A1) $m_b = 1 - e^{-\lambda_b W_b} \approx 1$, stopping "at a find" gives $m_b = 0$, and stopping "after a drought" gives $m_b > 1$. If (A1) is violated, m_b can be any non-negative number.

Only the *upper* bound survives rigorously. If the frontier is a stopping time with respect to the search history (the decision to stop does not look ahead), then by Wald's identity for the Poisson process $\mathbb{E}[N_b] = \lambda_b\, \mathbb{E}[W_b]$ *under any* stopping rule with $\mathbb{E}[W_b] < \infty$. Since $W_b = D_b + \delta_b \geq D_b$, we get $\mathbb{E}[M] \geq \kappa\, \mathbb{E}[S]$: in the sense of moments M/S bounds κ from above. There is no assumption-free lower bound. The rule "stop the search after a stretch of g mean intervals without finds" gives $\lambda_b \delta_b \geq g$. Moreover, waiting for such a stretch itself packs the events before D_b more densely, so that law (12) no longer holds. Table 3 shows the consequences: at $g = 2$ the estimator $(M-B)/S$ is inflated by 6.7%, and the exact test of Proposition 3 rejects a true hypothesis in 18% of cases instead of 5%. The rule "stop at the next find" ($\delta_b = 0$) biases $(M - B)/S$ downward by 6.4%, and then the estimator $(M - 1)/S$ is the correct one.

The precise statement is therefore as follows. Under (A1) the main result (20) is exact. Under stopping "at a find" the estimator (21) is correct. An upper bound in the sense of moments under any rule is given by M/S (Wald's identity); it differs from (21) only by $1/S$, that is, by about 0.5%, but the estimator (21) itself is not such a bound: under stopping "at a find" its mean is $0.996\,\kappa$ (Table 3). Under stopping "after a drought" the true κ is *below* $(M - B)/S$, that is, even closer to e^γ or below it. None of the violations of (A1) creates an excess over LPW that the data would require explaining. But without documented frontiers the size of the uncertainty is not bounded from below: this is a stronger argument for publishing L_b than the one given previously.

The unknown stopping rule nevertheless reduces to a single number. Put $m = \sum_b m_b / B$. By Wald's identity $\mathbb{E}[M] = \kappa\, \mathbb{E}[S] + Bm$ under any rule, so $(M - Bm)/S$ is unbiased in the sense of expectations. The two right-hand columns of Table 3 show that with m known this also holds for the means of the ratio (0.999–1.001), and that the level of the Poisson test for $M - Bm$ stays close to nominal, 0.043–0.056, under every rule, including drought $g = 2$, where test (13) errs in 18% of cases. A partial violation of (A1)—stopping "at a find" for only a fraction q of the bases—shifts $(M - B)/S$ downward almost linearly in q, by 1.8%, 3.4% and 4.9% at $q = 0.25$, 0.5 and 0.75, and raises the level of test (13) to 0.056, 0.080 and 0.117. Therefore §5.2 states the inference as a function of m (Table 6).

4.4 Calibration of the old pivot

How bad the Gamma pivot (10) is when the data are generated by scheme B depends on how far the frontier has advanced—and this should be reported as a range rather than as a single number. Table 4 gives both quantities at once: the type I error of the old test and the relative bias of both pooled estimators.

Three conclusions follow from Table 4. First, under scheme B a nominal 5% test rejects a true hypothesis in 11–17% of cases (in the table 0.114–0.163; for the first row with $6 \cdot 10^5$ replications 0.167 ± 0.0005, tag [S29]): the claimed precision of the intervals is overstated two- to threefold, and the more so the closer the frontier is to the last find. Second, the estimator $(M - 1)/S$ is biased upward by 4–7% under all rules, whereas $(M - B)/S$ is unbiased up to Monte Carlo noise under all rules—as Proposition 3 predicts, since conditional unbiasedness implies unconditional unbiasedness. Third, the figure "16.2%" quoted in an earlier version is the value for one particular scenario; the correct thing is to report the range. All seven rules of Table 4 place the frontier without regard to the finds, that is, they satisfy (A1); rules that violate (A1) are analyzed in Remark 10 and Table 3.

Table 3. Scheme B under various stopping rules: the mean gap $\bar{m} = \langle \lambda_b \delta_b \rangle$; the relative means of the pooled estimators $r_1 = \mathbb{E}[(M-1)/S]/\kappa$ and $r_B = \mathbb{E}[(M-B)/S]/\kappa$; the ratio $r_W = \mathbb{E}[M]/(\kappa \mathbb{E}[S])$ (at least 1 under every rule by Wald's identity); the type I error α_B of the exact two-sided Poisson test (13) at the 5% level; the same quantities for the estimator $(M - B\bar{m})/S$ and the test shifted by $B\bar{m}$ events, r_m and α_m. Budget $W_b^0 = N_b/\lambda_b$; "drought g": after the budget is spent the search continues until g/λ_b passes without a find; "mixture q": each base stops at the next find with probability q and at W_b^0 otherwise. 20 000 replications per row, accuracy about $\pm 0.05\%$ for the means and ± 0.003 for the level (script `stopping_rules.py`, tag [S13], seed 20260924).

Stopping rule	$\bar{m}$	r_1	r_B	r_W	α_B	r_m	α_m
(A1): frontier $= W_b^0$	0.996	1.0688	0.9999	1.0737	0.046	1.0002	0.046
(A1): frontier $= UW_b^0$, $U \sim \mathrm{U}(0.5, 2)$	0.997	1.0551	1.0000	1.0582	0.043	1.0002	0.043
stop at a find	0.000	0.9963	0.9362	1.0003	0.165	1.0006	0.045
drought $g = 0.5$	1.105	1.0762	1.0074	1.0811	0.050	0.9997	0.050
drought $g = 1$	1.366	1.0923	1.0250	1.0973	0.063	0.9987	0.050
drought $g = 2$	2.136	1.1227	1.0667	1.1277	0.178	0.9986	0.056
mixture $q = 0.25$	0.749	1.0489	0.9824	1.0534	0.056	1.0003	0.045
mixture $q = 0.5$	0.499	1.0304	0.9661	1.0346	0.080	1.0006	0.047
mixture $q = 0.75$	0.250	1.0127	0.9506	1.0168	0.117	1.0005	0.049
mixture with drought $g = 1$, $q = 0.5$	1.183	1.0821	1.0139	1.0870	0.053	1.0006	0.050

4.5 Documented frontiers and the check of (A1)

A dated and attributed frontier above the last find could be found for four bases (checked on 24 September 2026):

- $b = 2$: the GIMPS report at the time the data were read, $X_{\mathrm{ver}} = 81\,648\,221$ and $X_{\mathrm{ft}} = 141\,308\,443$ (§2);
- $b = 10$: "Search limit is 10800000, currently" (S. Batalov, A004023, 1 July 2021);
- $b = 22$ and $b = 26$: "No other terms < 100000" (R. Price, A127997, A127999, 2012; the last finds, 27 823 and 26 717, were made in 2007–2008).

The other mentions of search limits are either outdated and lie far below $n_{\max}$, or coincide with the last find and mark it rather than the search.

Under (A1) the quantities $\lambda_b \delta_b$ are independent and distributed as Exp(1) (the truncation at W_b is negligible here), so their sum over k bases has the Gamma$(k, 1)$ distribution. Small values point to stopping "at a find" ($m < 1$), large ones to stopping "after a drought" ($m > 1$, Remark 10). The observed values are 0.143 ($b = 2$ with X_{ver}; 0.093 with X_{ft}), 0.215 ($b = 10$), 0.737 ($b = 22$) and 0.722 ($b = 26$). Their mean is 0.44–0.45 against an expected value of one. The probability of so small a sum under (A1) is 0.10–0.11 (tag [S18]). These values are computed at $\lambda_b = e^\gamma / \ln b$, that is, under H_0: $\kappa = e^\gamma$; at $\kappa = \hat{\kappa}_{\mathrm{B}}$ every $\lambda_b \delta_b$ is larger by the factor 1.032, and the probability is 0.11–0.12 (tag [S29]); the conclusion does not change.

These numbers cannot, however, be read as a check of (A1) for our data snapshot, for three reasons. First, the frontiers for $b = 10$ (2021) and for $b = 22$ and $b = 26$ (2012) predate the snapshot of 27 August 2026. If the search continued without finds after publication, the true gap at the snapshot date exceeds the published one, so the observed $\lambda_b \delta_b$ are only lower bounds. Small published values are therefore no evidence of stopping at a find as of the snapshot, the lower tail 0.10 does not apply to it, and such values do not prove the absence of a "drought": the true gap may be arbitrarily large. Second, the moment at which a frontier is published may be tied to a find (for $b = 10$ it was reported two months after the find); by Remark 9 this does not

Table 4. Scheme B under seven frontier rules: type I error of the nominal 5% test built on the Gamma pivot (10), and the relative bias of the two pooled estimators. The rules are stated through the window width $W_b = \ln L_b - t_0$: "$\mathbb{E}[\#]$ = observed" is $W_b = N_b/\lambda_b$; "$+s$ mean intervals" is $W_b = (N_b + s)/\lambda_b$; "$c$ times farther in t" is $W_b = c\,D_b$ with the observed D_b. 50 000 replications per row, accuracy about ± 0.003 for the level and $\pm 0.1\%$ for the means (tag [S10]).

Rule for the frontier	type I error	$\mathbb{E}[(M-1)/S]/\kappa$	$\mathbb{E}[(M-B)/S]/\kappa$
$\mathbb{E}[\#\text{events}]$ = observed	0.1634	1.0691	1.0001
frontier +0.5 mean intervals	0.1620	1.0667	1.0002
frontier +1 mean interval	0.1596	1.0640	0.9998
frontier +2 mean intervals	0.1527	1.0599	0.9998
frontier +5 mean intervals	0.1359	1.0506	1.0002
frontier 1.5 times farther in t	0.1342	1.0499	0.9999
frontier twice as far in t	0.1136	1.0371	1.0001
A (control)	0.0492	—	—

violate law (12), but it makes $\lambda_b\delta_b$ small. Third, the bases with a published frontier are not a random sample: frontiers are reported where a particular participant does so, and nothing can be inferred from them about the other bases. Only the GIMPS frontier for $b = 2$ is synchronous with the snapshot and not tied to a find; for it $\lambda_2\delta_2 = 0.143$ (0.093 with X_{ft}), and the probability of a value no larger under (A1) is 0.133 (0.089; tag [S30])—a single observation, consistent with (A1).

Thus the documented frontiers neither confirm nor refute (A1). We therefore state the conclusion that the excess is created by the observation scheme conditionally, under (A1), and the conclusion of agreement with LPW for the whole family of stopping rules (§5.2, Table 6). Recall the direction. Stopping "after a drought" ($m > 1$) only lowers the true κ (Remark 10): it creates no excess over LPW, and a rejection from below would require a mean gap exceeding 3.2 mean spacings (Table 6; drought $g = 2$ gives $\bar{m} = 2.14$). The attribution of the excess would be weakened by the opposite deviation, $m < 1$: then $\mathbb{E}[M - B] \le \kappa\,\mathbb{E}[S] \le \mathbb{E}[M]$, and in the sense of expectations κ lies between $(M - B)/S$ and M/S.

The documented frontiers can also be used directly. For bases with a known frontier the full likelihood applies: $N_b \sim \text{Poisson}(\lambda_b W_b)$ with all events and the full window. For the others the conditional law (12) applies. The sum is again exactly Poisson. A frontier known only as a lower bound will do: the window $(t_0, \ln L_b]$ has then been covered completely, and all that is required is that the published bound was not chosen by the finds. For $b = 10$ this is doubtful: the frontier was reported two months after a find. We therefore take as the main one the hybrid estimate with the known frontier of a single base, $b = 2$ (X_{ft}): $\hat{\kappa} = 1.846$ with an exact 95% interval [1.602, 2.118] and $p = 0.626$ against e^{γ}; adding $b = 10$ gives 1.853, [1.608, 2.125], $p = 0.588$ (tag [S18]). Both differ from $(M - B)/S = 1.838$ by less than 1%, and the main conclusion does not change.

One more control needs neither documented frontiers nor conditioning. Every $n_{\max}$ exceeds $T = 2 \cdot 10^5$, so under contiguous search the window $(n_{\min}, T]$ has been covered for all fifteen bases, its width is known, and the full Poisson likelihood with all events of the window applies. We obtain $M = 163$, $S = 86.233$, $\hat{\kappa} = M/S = 1.890$ with an exact 95% interval [1.611, 2.204] and $p = 0.468$ (tag [S29]). Condition (A1) is needed here only in a weak form: the decision to take the search up to $2 \cdot 10^5$ did not depend on the finds below that bound. The estimate lies between $(M - B)/S$ and $(M - 1)/S$ and is consistent with LPW.

4.6 The data do not choose the scheme for us

We stress that $(M-1)/S$ and $(M-B)/S$ are functions of the same sufficient statistics (N_b, D_b). In principle the schemes are distinguishable: under scheme A one expects $N_b \approx \kappa D_b/\ln b$, under scheme B one more. But a difference of one event per base drowns in the Poisson scatter of each base, so the finds themselves practically cannot choose the scheme: the choice is determined by how the data were collected, and its price is 6.9%—twice the entire residual gap with LPW (3.2%).

The natural attempt to sidestep the choice is to build the pooled estimate out of pieces that are unbiased by construction. The per-base estimator $\tilde{\kappa}_b = (N_b - 1)\ln b/D_b$ is unbiased under scheme A (Corollary 1) and unbiased under scheme B up to an exponentially small term. Indeed, for $N_b = n \geq 2$ the ratio D_b/W_b is distributed as $\mathrm{Beta}(n, 1)$, and the identity $\mathbb{E}[1/\mathrm{Beta}(\alpha,\beta)] = (\alpha+\beta-1)/(\alpha-1)$ gives $\mathbb{E}[\tilde{\lambda}_b \mid N_b = n] = n/W_b$; for $n \in \{0, 1\}$ the estimator is zero. Averaging over $n \sim \mathrm{Pois}(\lambda_b W_b)$ therefore gives not λ_b but

$$\mathbb{E}[\tilde{\lambda}_b] = \lambda_b - \frac{\Pr(N_b = 1)}{W_b} = \lambda_b\Big(1 - e^{-\lambda_b W_b}\Big),$$

that is, the same exponentially small deficit: the plug-in $\lambda_b W_b \approx N_b \geq 7$ puts it at 0.1% or less.

Nevertheless, the weighted mean

$$\kappa_* = \sum\nolimits_b (N_b - 2)\tilde{\kappa}_b \Big/ \sum\nolimits_b (N_b - 2)$$

with weights inversely proportional to the variance under scheme A ($\mathrm{Var}\,\tilde{\kappa}_b = \kappa^2/(N_b - 2)$; under scheme B the unconditional variance of $\tilde{\kappa}_b$ is infinite, since $\Pr(N_b = 2) > 0$ and for $N_b = 2$ we have $D_b/W_b \sim \mathrm{Beta}(2, 1)$ and $\mathbb{E}[D_b^{-2}] = \infty$) has a bias of 0.0% under scheme A but +7.5% under scheme B with the frontier $W_b = N_b/\lambda_b$ (+3.9% with the frontier $W_b = 2D_b$; tag [S30]), that is, practically the same as $(M-1)/S$ under the same rules (Table 4). The reason is that the weights—the event counts themselves—are random under scheme B and positively correlated with the estimates; unbiasedness of each summand does not carry over to a ratio of sums. The correct way to pool is not to weight the per-base estimates but to sum the sufficient statistics, as (13) does.

From this we draw a narrow but, we think, important conclusion: *the gain in precision from pooling the bases is inseparable from an assumption about the observation scheme.* The only radical remedy is to document the search frontiers L_b, as GIMPS does for $b = 2$. We address this request to the participants of the relevant search projects.

5 Results

5.1 Per-base estimates

Eleven of the fifteen bases lie below $e^{-\gamma}$ under the endpoint form, and eleven of the fifteen under OLS—but these are *different* sets of eleven. The membership changes: base 18 lies above $e^{-\gamma}$ under the endpoint form and below it under OLS, and base 20 the other way round. In other words, even the sign of the deviation is not fully robust to the choice of form; only the *number* of bases below $e^{-\gamma}$ is. This is the same effect as in §3.4, and it affects exactly the two bases for which the choice of form weighs more than the deviation itself.

All fifteen mid-PIT quantiles lie inside the central 95% region; the extremes are $b = 19$ at quantile 0.911 and $b = 7$ at 0.160. Base 13, the "lucky" one, sits at 0.780, which is in no way remarkable for one base out of fifteen. All fifteen Garwood intervals cover e^{γ}.

Table 5. Per-base estimates. $\hat{G}_b = D_b/(N_b \ln b)$ is the endpoint form; $\hat{G}_b^{\mathrm{OLS}}$ is the OLS slope of $\ln n_k$ on k over all known indices including $n_{\min}$ ($k = 0, \ldots, N_b$), divided by $\ln b$; both are unbiased (Proposition 2). $\hat{\kappa}_b = K_b/(D_b/\ln b)$ is the scheme B conditional estimator with the exact Garwood interval (15) from Poisson($\kappa D_b/\ln b$). The last column is the Poisson mid-PIT at $\kappa = e^{\gamma}$: values near one correspond to bases with a *high* κ_b (orientation opposite to the Gamma PIT of the earlier version). LPW predicts $G = e^{-\gamma} = 0.5615$ and $\kappa = e^{\gamma} = 1.7811$.

b	N_b	K_b	$\hat{G}_b$	$\hat{G}_b^{\mathrm{OLS}}$	diff., %	$\hat{\kappa}_b$	95% CI (Garwood)	mid-PIT
2	51	50	0.5102	0.5309	+4.1	1.921	[1.426, 2.533]	0.710
3	22	21	0.6148	0.6081	−1.1	1.552	[0.961, 2.373]	0.272
5	18	17	0.4802	0.4666	−2.8	1.967	[1.146, 3.149]	0.670
6	16	15	0.5000	0.4904	−1.9	1.875	[1.049, 3.092]	0.594
7	9	8	0.7104	0.7371	+3.8	1.251	[0.540, 2.466]	0.160
10	10	9	0.6612	0.6746	+2.0	1.361	[0.622, 2.584]	0.216
11	12	11	0.4034	0.3617	−10.3	2.272	[1.134, 4.066]	0.794
12	13	12	0.3981	0.3796	−4.7	2.319	[1.198, 4.050]	0.820
13	12	11	0.4098	0.3448	−15.9	2.237	[1.117, 4.002]	0.780
14	10	9	0.5025	0.5386	+7.2	1.791	[0.819, 3.400]	0.528
15	9	8	0.5035	0.5054	+0.4	1.766	[0.762, 3.479]	0.513
17	11	10	0.4301	0.4353	+1.2	2.114	[1.014, 3.887]	0.717
18	7	6	0.6604	0.4656	−29.5	1.298	[0.476, 2.825]	0.228
19	10	9	0.3161	0.2971	−6.0	2.847	[1.302, 5.405]	0.911
20	8	7	0.5300	0.5713	+7.8	1.651	[0.664, 3.402]	0.444

5.2 The pooled estimate

The exact conditional inference for scheme B (Proposition 3) gives

$$\hat{\kappa}_{\mathrm{B}} = \frac{M - B}{S} = 1.8381, \qquad 95\%\ \mathrm{CI} = [1.5939,\ 2.1091], \qquad p = 0.672, \tag{20}$$

where the interval is (15) with $K = 203$, $S = 110.4423$, and the p-value is obtained by the doubled-minimum-tail convention for $K \sim \mathrm{Poisson}(e^{\gamma}S)$, $e^{\gamma}S = 196.71$; under the mid-p convention $p = 0.647$. The exact coverage of interval (20) is 0.9540 at $\kappa = e^{\gamma}$ and does not fall below 0.950 for any κ.

For comparison, scheme A via (9)–(10) gives

$$\tilde{\kappa} = 1.9648, \qquad 95\%\ \mathrm{CI} = [1.7205,\ 2.2444], \qquad p = 0.142. \tag{21}$$

The point estimate (21) is 10.3% above e^{γ}, and it is precisely this excess that used to be interpreted as the trace of a factor omitted by the heuristic. On moving to scheme B the excess falls to 3.2%, and the p-value rises from 0.14 to 0.67.

We hold that scheme B is closer to the way the data were obtained, and therefore regard (20) as the main result under (A1). Estimate (21) corresponds to stopping the search exactly at a find; under any stopping rule κ is bounded from above in the sense of expectations by $M/S = 1.974$, only 0.5% larger than (21) (Remark 10). Under stopping "after a drought" the true value lies below (20).

Without (A1) the unknown stopping rule enters the pooled estimate through a single number—the mean gap m between the last find and the frontier in units of the mean spacing (Remark 10): $(M - Bm)/S$ is unbiased in the sense of expectations, and with m known the Poisson test for $M - Bm$ approximately keeps its level under every rule of Table 3. Table 6 gives the inference as a function of m. Under (A1) $m \approx 1$ and one obtains (20); stopping exactly at a find gives $m = 0$; stopping "after a drought" gives $m > 1$. LPW is not rejected for any m from 0 to 3.2; the estimate equals e^{γ} at $m = 1.42$. With the second-order term (§9) LPW is not

Table 6. The pooled estimate as a function of the unknown mean gap m between the last find and the frontier (in mean spacings): $(M - Bm)/S$, the excess over e^γ, the Garwood interval and the exact two-sided p for $K_m = M - Bm \sim \text{Poisson}(\kappa S)$ in the constant-rate model; the two right columns give the same with the calibrated second-order term and the exposure $S_\Sigma(c)$ (§9). $m = 1$ is (A1), $m = 0$ is stopping exactly at a find, $m > 1$ is stopping after a drought. The shifted test is exact under (A1) and, with m known, approximately keeps its level (0.043–0.056) under every rule of Table 3. Tag [S30].

m	$(M - Bm)/S$	excess, %	95% CI (Garwood)	p	$K_m/S_\Sigma(c)$	p
0 (at a find)	1.974	+10.8	[1.721, 2.254]	0.142	1.911	0.314
0.4	1.920	+7.8	[1.670, 2.196]	0.292	1.859	0.553
1 (A1)	1.838	+3.2	[1.594, 2.109]	0.672	1.780	1.000
1.4	1.784	+0.1	[1.543, 2.051]	1.000	1.727	0.699
2	1.702	−4.4	[1.468, 1.964]	0.564	1.648	0.303
3	1.566	−12.1	[1.342, 1.818]	0.094	1.517	0.034

rejected for m from 0 to 2.8, and equality with e^γ is reached at $m = 0.99$, that is, exactly under (A1). Thus the conclusion of agreement with LPW is robust to violations of (A1): it holds for every stopping rule with a mean gap up to 3.2 (2.8) mean spacings, drought $g = 2$ included, whereas the conclusion that the excess is created by the observation scheme does depend on it: at $m = 0$ the residual excess is 10.8%. (A1) cannot be checked from the finds themselves, and the documented frontiers do not settle it (§4.5). Under (A1) the remainder is moreover explained by the second-order term of the heuristic itself (§9).

The overall picture is shown in Figure 1: the confidence interval of every base covers e^γ as well as both pooled estimates.

5.3 The homogeneity test and its power

Let H_0 be the model with a single κ and H_1 the model with free κ_b. From the conditional likelihood of Proposition 3, since $\sum_b \hat\kappa_b D_b/\ln b = K = \hat\kappa_{\mathrm{B}} S$, the likelihood-ratio statistic reduces to

$$\Lambda = 2\sum_b K_b \ln\frac{\hat\kappa_b}{\hat\kappa_{\mathrm{B}}} = 2\sum_b K_b \ln\frac{K_b}{K\, s_b/S}, \qquad s_b = \frac{D_b}{\ln b}. \tag{22}$$

Crucially, its null distribution is *exact* here: conditioning on $K = \sum_b K_b$ turns (13) into the multinomial model $(K_b) \mid K \sim \text{Multinomial}(K;\, s_b/S)$, and there is no need to appeal to χ^2_{B-1} asymptotics. We obtain $\Lambda = 7.127$, exact $p = 0.935$ ($2 \cdot 10^6$ multinomial replications, ± 0.0003 at the 95% level); the asymptotic approximation gives $p = 0.930$ on 14 degrees of freedom, that is, it works well in this case. An exact null law exists under scheme A as well: there, under H_0, the quantities κs_b are independent $\text{Gamma}(N_b, 1)$ variables, so the vector s_b/S is Dirichlet with parameters N_b and does not depend on κ; the analogue of statistic (22) with N_b in place of K_b is $\Lambda_{\mathrm{A}} = 7.395$ with exact $p = 0.923$ (χ^2: 0.918; tag [S30]).

But a non-rejection is informative exactly to the extent that the test is powerful, and here one must apply to oneself the same requirement one applies to the literature. We simulated alternatives in which the κ_b are lognormally scattered with mean e^γ and coefficient of variation cv (that is, $\ln\kappa_b$ has variance $\ln(1 + cv^2)$), conditionally on the observed K: $(K_b) \mid K \sim \text{Multinomial}\big(K;\, \kappa_b s_b/\sum_{b'} \kappa_{b'} s_{b'}\big)$; the critical value is the 95% quantile of the exact null law:

cv	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50
power	0.059	0.097	0.184	0.326	0.484	0.645	0.774	0.851	0.948

The test attains 80% power only at $cv \approx 0.37$ (12 000 replications per grid point, power accurate to about ± 0.004). The centre of the distribution of κ_b (mean or median) does not

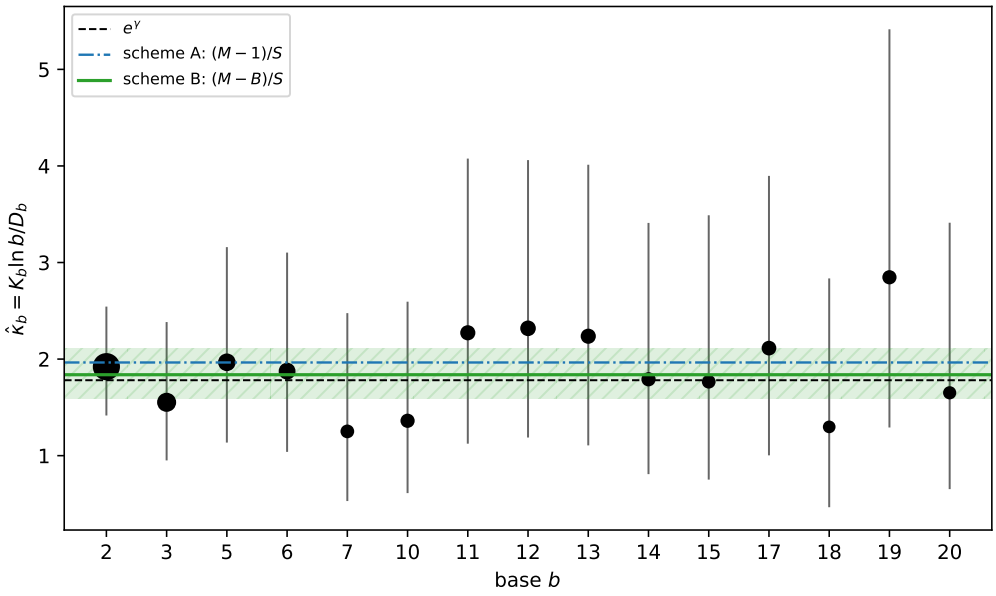


Figure 1. Per-base estimates $\hat{\kappa}_b$ with exact 95% Garwood intervals (15); marker area is proportional to N_b. Dashed line: the LPW value e^γ; green with a band: the pooled scheme B estimate and its interval; blue dash-dotted: the scheme A estimate $(M-1)/S$. Each base separately is compatible with a universal constant.

affect the power: the multinomial probabilities $\kappa_b s_b / \sum_{b'} \kappa_{b'} s_{b'}$ do not change when all κ_b are multiplied by the same number, and only the spread matters. Hence $p = 0.935$ means: the data do not require base-dependent frequencies, but neither do they exclude a spread of κ_b with a coefficient of variation of up to a third. The statement "lucky bases do not exist" does not follow from the available data; what follows is the weaker statement "their existence is not supported by the data, and a spread with a coefficient of variation below 0.37 is detected with probability below 80%".

5.4 Robustness

Lower truncation. We varied the lower truncation point in order to probe the asymptotic regime (the heuristic (2) is asymptotic, whereas the constant-intensity model starts at $n = 2$). The results are in Table 7. The scheme A estimate grows from 1.965 to 2.085 in the strictly consistent variant $N_b \geq 1$ (to 2.099 with the selection $N_b \geq 2$)—not strictly monotonically (at $n > 10^2$ it retreats from 2.019 to 2.012) but systematically: the growth is 6.1% (6.8%), and all values come from nested subsamples of the same data. It is not, however, a significant trend: simulating the data on the same origins and event counts (under scheme B with the frontier $W_b = N_b/\lambda_b$, the rule "$\mathbb{E}[\#]$ = observed" of Table 4), a rise of the scheme A estimate by at least 6.1% occurs with probability 0.219 under scheme A and 0.615 under scheme B (with the selection $N_b \geq 2$, for a rise of 6.8%: 0.196 and 0.597; tags [S29], [S30]). The scheme B estimate shows no upward trend and stays close to e^γ, within 1.757–1.871 in the strictly consistent variant $N_b \geq 1$ (with the selection $N_b \geq 2$, 1.789–1.871), and its p-values lie between 0.53 and 1.00 at all truncation levels. The reason is visible from (13): truncation reduces M but not B, whereas scheme A subtracts one regardless of the sample size; the difference of the two estimates is $(B-1)/S$ and grows mechanically, since S falls under truncation from about 110 to 42. The truncation profile is therefore consistent with the diagnosis of §4 and weakly favours scheme B (a likelihood ratio of about three), but it is not an independent confirmation. From $n > 10$ onwards the scheme B estimate decreases (1.871 → 1.757, by 6.1%, in the strictly consistent

Table 7. Robustness to lower truncation. A base is retained if at least *two* events survive the truncation ($N_b \geq 2$); at $n > 10^4$ base 19 no longer meets this criterion, whence $B = 14$. The criterion matters: under the weaker requirement $N_b \geq 1$ base 19 is retained, giving the last row; only that row is strictly consistent with Proposition 3, since selecting on $N_b \geq 2$ means selecting on $K_b \geq 1$ and biases the estimate upward (Remark 1). The interval and p_{B} are exact, scheme B; the last column (p_{A}, the scheme A Gamma pivot) is given for comparison with the old practice.

truncation	B	M	$(M-1)/S$	$(M-B)/S$	95% CI, scheme B	p_{B}	p_{A}
none	15	218	1.9648	1.8381	[1.594, 2.109]	0.672	0.142
$n > 10$	15	192	2.0192	1.8712	[1.606, 2.168]	0.531	0.080
$n > 10^2$	15	156	2.0116	1.8299	[1.540, 2.158]	0.771	0.124
$n > 10^3$	15	125	2.0551	1.8231	[1.498, 2.197]	0.832	0.106
$n > 10^4$	14	89	2.0990	1.7889	[1.407, 2.242]	1.000	0.116
$n > 10^4$, $N_b \geq 1$	15	90	2.0848	1.7569	[1.382, 2.202]	0.966	0.128

variant; to 1.789 with the selection $N_b \geq 2$); this too is within the noise of nested subsamples: under scheme B with the same frontier a fall from $n > 10$ to $n > 10^4$ at least as large as the observed one occurs with probability 0.229, and a fall from the full sample to $n > 10^4$ at least as large as the observed one (4.4%) with probability 0.306 (with the selection $N_b \geq 2$, for a fall of 2.7%, 0.308; tags [S29], [S30]). The second-order term does not remove this decrease (§9). The curves are shown in Figure 2.

The result does not depend on where the origin is placed either. With the origin at the truncation point itself (the first surviving event is then not lost), the scheme B estimate is 1.891, 1.867, 1.906 and 1.891 at $n > 10$, 10^2, 10^3, 10^4 ($p = 0.43$–0.60). If for the five bases whose origin $n_{\min} = 2$ is fixed by arithmetic (§2.2) the origin is moved to the second index, then $(M-B)/S = 1.882$ with $p = 0.457$ (tag [S29]).

Leave-one-base-out. A leave-one-base-out analysis gives $\hat{\kappa}_{\mathrm{B}}$ within [1.808, 1.878] (the extreme values arise when $b = 19$ and $b = 3$ are removed) and exact p-values within [0.49, 0.85]; no single base determines the conclusion.

Status of the data. Restricting base 2 to the fifty double-checked exponents moves the scheme B estimate from 1.838 to 1.834 (scheme A: from $\tilde{\kappa} = 1.965$, $p = 0.142$ to $\tilde{\kappa} = 1.961$, $p = 0.151$). PRP status begins far below one million for most terms (§2), so the next variant tests not PRP status as such but the most recent and least re-checked finds: excluding all indices $n > 10^6$ gives $M = 182$ and $(M-B)/S = 1.800$ ($\tilde{\kappa} = 1.951$). In the worst case, when the last known PRP of some base turns out to be composite, the largest shift comes from $b = 17$: $(M-B)/S = 1.848$ ($\tilde{\kappa} = 1.977$). All variants lie inside both interval (21) and interval (20), so possible pseudoprimes do not affect the conclusion. More important than spurious finds are missed ones (§2): a miss lowers K at unchanged S and biases the estimate downward, in the direction of the main conclusion. For LPW to be rejected from above, at least 23 finds would have to be missing inside the windows, 1.5 per base, and at least 30 with the second-order term (tag [S30]). The two 2024 finds for base 5 do not fall inside the windows: they lie above $n_{\max}$, beyond an unsearched gap (§2).

6 Statistical power

By (13) the power against a true constant $\kappa = re^{\gamma}$ at level α is available in closed form (Table 8). The convention has to be stated: throughout this work the test is *two-sided*, with the doubled-minimum-tail convention, $\alpha = 0.05$. The one-sided variant of the same test gives noticeably

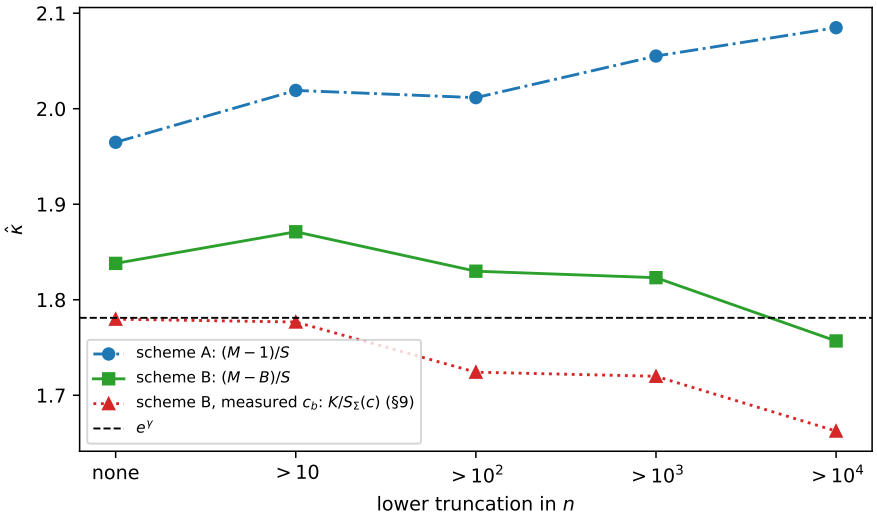


Figure 2. The pooled estimate under lower truncation: $(M-1)/S$ (scheme A), $(M-B)/S$ (scheme B) and $K/S_\Sigma(c)$ with the measured second-order term and the exposure summed over the primes (§9). The last point ($n > 10^4$) is the strictly consistent variant $N_b \geq 1$ (Table 7). The upward trend of scheme A disappears under scheme B. All curves come from nested subsamples, and their changes under truncation are within the noise (§5.4).

Table 8. Power at level $\alpha = 0.05$ for rejecting $\kappa = e^\gamma$ when the true constant exceeds it by the stated factor. Exposures: $S_2 = 26.02$ ($K = 50$) for base 2 alone and $S = 110.44$ ($K = 203$) for the pooled sample.

	two-sided		one-sided	
κ/e^γ	Mersenne only	pooled	Mersenne only	pooled
1.05	0.060	0.099	0.082	0.165
1.10	0.099	0.266	0.147	0.386
1.15	0.164	0.514	0.235	0.644
1.20	0.253	0.752	0.343	0.844
1.30	0.479	0.972	0.581	0.988
1.50	0.861	1.000	0.910	1.000

different numbers; they are shown in the right-hand part of the table for comparability with works in which the convention is not stated.

At 80% power the Mersenne sequence alone detects only deviations of 45.5% and above; pooling fifteen bases lowers that threshold to 21.3% (one-sided variant: to 18.7%). Table 8 concerns an excess; for a deficit of the constant the thresholds of the same two-sided test differ: 37.4% for base 2 alone and 19.2% for the pooled sample (tag [S29]). Attaining 80% power against a 10% deviation requires an exposure $S \approx 470$, that is, $K \approx 837$ conditional events ($M \approx 852$) at $\kappa = e^\gamma$ (if the true constant is 10% larger, one expects $K \approx 921$ conditional events, that is, $M \approx 936$ finds); against 5%, $S \approx 1823$ and $K \approx 3246$. The power function of a discrete test is not monotone in S (it is saw-toothed), so the values given are those beyond which the power no longer drops below 80% (found by an exhaustive scan over S with step 0.01).

These thresholds are computationally out of reach, and it is worth saying by how much. Since K grows logarithmically in the search frontier, doubling the frontiers *of all fifteen bases* adds $\ln 2 \sum_b 1/\ln b = 5.41$ to the exposure, that is, on average $e^\gamma \cdot 5.41 = 9.64$ conditional events. To raise the exposure from 110.4 to 470 one needs $(470 - 110.4)/5.41 \approx 66.4$ doublings, that is, a growth of the frontiers by a factor of about $2^{66.4} \approx 10^{20}$; for a sensitivity of 5%, about 316

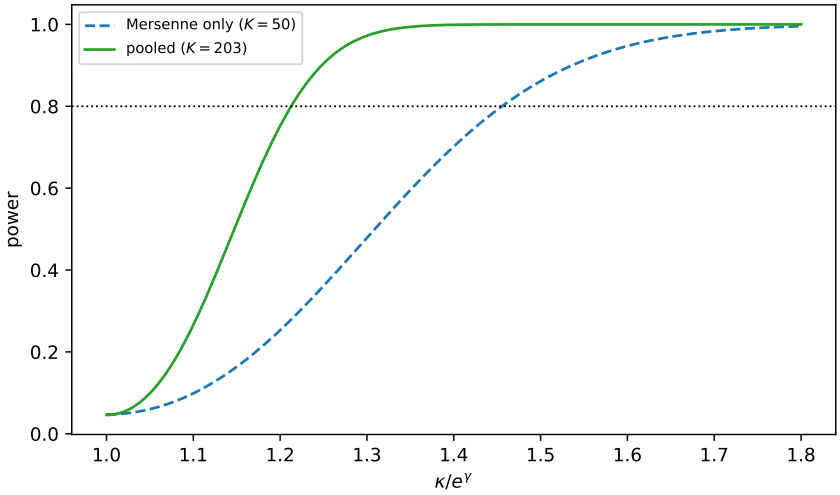


Figure 3. Power of the two-sided rejection of $\kappa = e^{\gamma}$ at level $\alpha = 0.05$. The horizontal line marks 80% power.

doublings, that is, a factor of order 10^{95}. This is not a matter of decades but of magnitudes lying beyond the reach of any conceivable computation. Both power curves, as functions of the true deviation, are shown in Figure 3.

This yields a rule for reading this literature. A report that the data agree with LPW to within a few percent is not evidence for LPW at the level of a few percent; it is compatible with a constant that is wrong by a factor of one and a half. The pooled analysis does, however, establish a genuine bound: the universal constant, if it exists, lies under (A1) roughly in [1.59, 2.11] in the constant-intensity model and in [1.52, 2.06] with the measured second-order term at $n > 10$ (§9); no base deviates from it detectably.

7 The same error in a different guise

The Mersenne counting function invites a related error, which we present as a worked example. Regressing $\pi_2(t)$ on t at the event points, over the 50 double-checked exponents (numbered $0, \dots, N-1$: the smallest exponent is the origin, not an event), gives a slope of 2.6391 and an intercept of -2.9627 with an OLS standard error of 0.4469—apparently significant, and suggestive of a secondary term in (2). Nothing of the sort.

Two defects compound. The residuals of the cumulative count are strongly autocorrelated, here $\rho_1 = 0.825$ (the sample correlation of adjacent residuals; the sample autocorrelation function gives 0.751); the Newey–West estimator with the Bartlett kernel and eight lags, without a degrees-of-freedom correction, raises the standard error to 0.7315 (with the correction $n/(n-k)$, to 0.7466; tag [S30]). More importantly, the regression uses only the event points and thereby discards the empty interval between the last exponent and the search frontier. The correct null model holds the interval $[t_0, T]$ fixed, where $T = \ln X_{\mathrm{ver}}$, and conditions on exactly $N - 1 = 49$ events strictly to the right of t_0, which, by the order-statistics property of the Poisson process given the number of events (cf. the proof of Proposition 2(b)), means 49 independent uniform points on $(t_0, T]$. Over 60 000 such replications the intercept has median -1.529 and a 5–95% range of $[-5.99, 2.20]$, giving a two-sided $p = 0.570$ for the observed value. The secondary term is an artifact of conditioning. Note that even the correct null distribution is not centred at zero. The count vanishes at $t_0 = \ln 2$, while the intercept is measured at $t = 0$, and this alone shifts it by about $-\hat{\lambda} \ln 2$, that is, by almost -1.9. Comparing -2.96 with zero is thus doubly wrong: because of the understated standard error, and because the null expectation is not zero.

7.1 Three consequences of one and the same conditioning

(i) The bootstrap does not help. The natural reaction to a suspiciously small OLS standard error is a nonparametric pairs bootstrap over the event points. On the base-2 data, a bootstrap over 20 000 resamples gives median -2.948, standard deviation 0.385 and a 95% interval $[-3.809, -2.300]$ lying entirely below zero; the proportion of negative values is 100%. This is not a confirmation but a repetition of the same error: resampling the observed points reproduces the variability within the realized configuration and cannot recover the variability of the process that produced that configuration. The correct null distribution has standard deviation 2.499 (5.6 times the OLS standard error of 0.447), that is, it is 6.5 times wider than the bootstrap one *in terms of standard deviation* (and likewise 6.5 times in terms of the width of the 95% interval).

(ii) Spurious underdispersion. Comparing the observed $\max_i |r_i| = 6.347$, where $r_i = i - \hat{\lambda}(t_i - t_0)$, with a simulation of an *unconditioned* Poisson process suggests the conclusion that the counting function is "smoother than Poisson". Against the correct null distribution—conditioning on $N-1$ events and estimating λ from the last of them—the median is 5.117 and the two-sided significance level is 0.529. The observation is entirely ordinary. An unnormalized supremum is insensitive near the pinned ends, so we also checked the version standardized by the bridge variance, $\max_i |r_i| / \sqrt{(N-1)u_i(1-u_i)}$ with $u_i = (t_i - t_0)/(t_{N-1} - t_0)$ over the interior events, under the same null law: it gives 3.45 with $p = 0.194$, again an ordinary value (tag [S29]).

(iii) The "mid-range deficit". Estimating λ from the last event forces the residual r to vanish at both ends ($r(t_0) = r(t_N) = 0$ under the correct numbering), turning its trajectory into an analogue of a Brownian bridge. Pinning the ends sets the typical size of an excursion in the middle of the range, but not its sign: the bridge is asymptotically symmetric (with $N - 1 = 49$ the asymmetry is negligible), and a negative excursion is no more likely than a positive one of the same size. The observed mean of the residuals r_i at the events in the middle half of the t-axis, that is, with $u_i \in [1/4, 3/4]$, is -3.13 against a simulation median of -0.012 and $p = 0.272$; the proportion of replications with a negative mean excursion is 0.502, so there is no systematic deficit. The trajectory of the residual process itself, together with the simulation band, is shown in Figure 4.

7.2 Base 2 at a known frontier

Base 2 is one of the two bases of the main set with a published frontier (the other is $b = 10$, §4.5); for it a scheme B analysis is possible *using the full exposure* rather than only the conditional one. With $T = \ln X_{\mathrm{ver}}$ we obtain $\hat{\lambda} = (N-1)/(T - t_0) = 2.7960$ against $e^{\gamma}/\ln 2 = 2.5695$, that is, $\hat{\kappa}_2 = 1.9381$ with an exact 95% Garwood interval [1.434, 2.562] and a Poisson p-value of 0.593 (doubled tail; under the mid-p convention, 0.545—the convention must be stated here, since the difference is appreciable). The conditional estimate on *the same slice of data*—the same 50 double-checked exponents—gives 1.9046; the difference is small because the last exponent of that slice, 77 232 917, happens to lie very close to the frontier 81 648 221. (For comparison, the conditional estimate over all 52 known exponents, reported in Table 5, is 1.9215; it must not be compared with $\hat{\kappa}_2$, being a different sample.) This is a coincidence of one realization, not a property of the method: the correction of §4.3 concerns the average over the position of the last event relative to the frontier.

We note in passing that the regime described here is exactly the one to which we urge a transition for all bases: with W_b known, conditioning on D_b becomes unnecessary, and the precision increases by the very 3.7% discussed in §4.2.

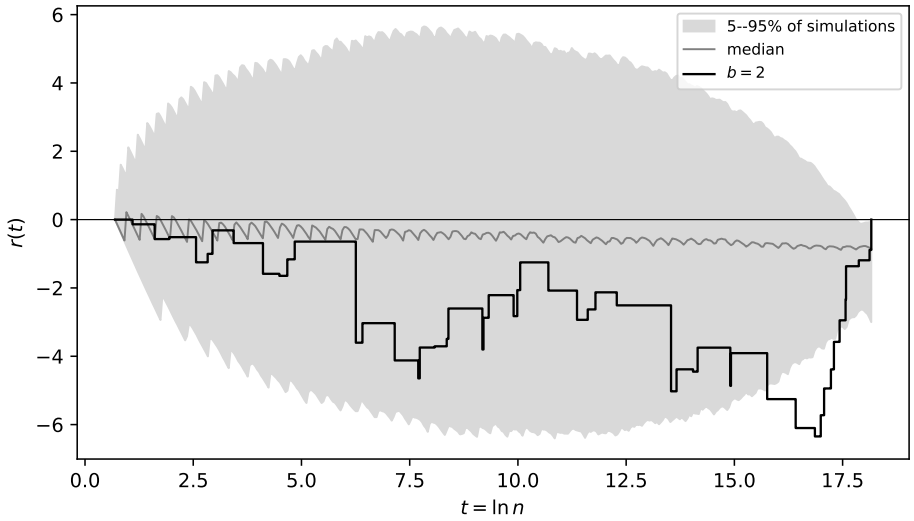


Figure 4. The residual process $r(t)$ for base $b = 2$ together with the pointwise 5–95% quantile band and the median of the simulations. The band is built for the same step function $r(t) = \#\{t_i \le t\} - \hat{\lambda}(t - t_0)$ under the null law of §7: 49 uniform points on $(t_0, \ln X_{\mathrm{ver}}]$, $\hat{\lambda}$ from the last of them. The trajectory leaves the band near $t \approx 16.9$, where $r = -6.35$ against a pointwise 5% quantile of -3.85 (the band drawn; an independent simulation [S29] gives -3.81). A pointwise exit says nothing about significance: for the whole trajectory the standardized supremum gives $p = 0.194$, the plain one $p = 0.529$ (§7, tag [S29]).

7.3 The sign of the intercept proves nothing

The previous version gave here an "independent argument": that the theoretical secondary term, $(e^{\gamma}/\ln 2)\mathfrak{M} = +0.672$ with the Meissel–Mertens constant $\mathfrak{M} = 0.26150\ldots$, has the sign opposite to the observed -2.963. This argument is wrong, and we withdraw it: it takes into account only one of the two Mertens theorems. Summing (24) over primes $p \le x$ with $\sum_{p \le x} \ln p/p = \ln x + E + o(1)$, $E = -1.3326\ldots$, and $\sum_{p \le x} 1/p = \ln\ln x + \mathfrak{M} + o(1)$ gives

$$\pi_2(x) \approx \frac{e^{\gamma}}{\ln 2}\bigl(\ln x + c_2 \ln\ln x\bigr) + C_0, \qquad C_0 = \frac{e^{\gamma}}{\ln 2}\bigl(E + c_2\mathfrak{M}\bigr).$$

With the measured $c_2 = 0.96$ (§9) we get $C_0 = -2.78$; direct summation over $p \le 10^8$ gives the same value. The constant C_0 itself depends strongly on the treatment of small p, where probability (24) exceeds one: with the probabilities capped at one it becomes $C_0 = -5.05$ (tag [S30]). The intercept of the regression is even less C_0: it depends in addition on the numbering of events, on the linearization of the term $c_2 \ln\ln x$ and on the conditioning on the number of events. A direct check is more reliable. Let the heuristic itself generate the Mersenne exponents—Bernoulli trials over the primes $p \le X_{\mathrm{ver}}$ with probability (24) at the measured c_2—and condition on 50 events. The intercept of the same regression then has median -2.52 and a 5–95% range of $[-5.20, -0.15]$, and the observed -2.963 lies at its quantile 0.385; with Wagstaff's refinement ($a = 2$ or 6) the median is -2.32 and the quantile 0.343 (tag [S28]). The second-order heuristic predicts a negative intercept of the same size as the observed one. What bears on the structure is not the sign of C_1 but only the correct null distribution (§7, $p = 0.57$): the intercept C_1 is a parameter of the fit and calls for no substantive interpretation.

Table 9. Comparison of structural models under the conditional Poisson likelihood of Proposition 3 (full form, including $-\ln K_b!$); LR and p are relative to model I; BIC is computed with $n = K = 203$.

Model	parameters	$\ln L$	LR	p	AIC	BIC	$P(\text{model} \mid \text{data})$
I	1	−35.753	—	—	73.51	76.82	0.906
II	3	−35.162	1.18	0.554	76.32	86.26	0.008
III	2	−35.458	0.59	0.443	74.92	81.54	0.085
IV	4	−34.797	1.91	0.591	77.59	90.85	0.001

8 Structural corrections

Consider the more general parametrization of the intensity $\lambda_b = (\kappa/\ln b)\, C_b$ and four models for the factor:

$$\begin{aligned}
\text{I:}\quad & C_b = 1;\\
\text{II:}\quad & C_b = \prod_{p\mid b-1,\ p\neq 2} \left(\tfrac{p}{p-1}\right)^{\alpha} \prod_{p\mid b+1,\ p\neq 2} \left(\tfrac{p}{p-2}\right)^{\beta};\\
\text{III:}\quad & C_b = \prod_{p\mid b^2-1} \left(\tfrac{p}{p-1}\right)^{\eta};\\
\text{IV:}\quad & C_b = \exp\bigl(\theta_1\omega(b-1) + \theta_2\omega(b+1) + \theta_3 \ln b\bigr).
\end{aligned}$$

The models are numbered with Roman numerals so as not to confuse them with the observation schemes A and B; the exponent in model III is written η so as not to confuse it with the Euler constant or with the distance to the frontier δ_b. The likelihood here is the *conditional Poisson* likelihood of Proposition 3, $\ell_{\mathrm{B}} = \sum_b \bigl[K_b \ln \mu_b - \mu_b - \ln K_b!\bigr]$ with $\mu_b = \kappa C_b D_b/\ln b$; the previous version used the scheme A likelihood (8), which is incompatible with §4. The results are in Table 9. No parametrization gives a significant improvement; the BIC-based Bayes factors at $n = K = 203$ strongly support the base model against models II and IV (112 and 1112) and moderately against III (10.6; at $n = 15$, 2.9; tag [S30]).

Models II–IV formalize the folklore notion of "lucky" bases through the divisors of $b \pm 1$, after the pattern of a Hardy–Littlewood singular series. Theory predicts no such factor: for an odd prime $p \nmid b-1$ the primes $q \mid b-1$ and $q \mid b+1$ do not divide $R_b(p)$, since $R_b(p) \equiv p$ and $R_b(p) \equiv 1 \pmod q$ respectively, but all primes $q < 2p+1$ behave in the same way for every b (§2), and this gain is already part of the common factor $e^{\gamma}\ln(2p)$. The heuristic predicts $C_b \equiv 1$, so this section tests the folklore rather than the theory; the substantive second-order dependence on the base is analysed in §9.

Remark 11 (Sensitivity of the BIC). The choice of effective sample size for the BIC is not obvious: the likelihood has $B = 15$ independent summands, but the information grows as $K = 203$. With $n = 15$ the posterior probabilities are 0.661 / 0.080 / 0.229 / 0.030; with $n = 203$ they are 0.906 / 0.008 / 0.085 / 0.001. The conclusion favors model I under either choice, but with $n = 203$ it is substantially stronger. We report $n = K$ as matching the number of independent conditional events.

A fixed arithmetic correction. We considered separately a model with a *fixed* (not estimated) factor $C_b^{\mathrm{fix}} = \prod_{p\mid b-1,p\neq 2} \frac{p}{p-1} \prod_{p\mid b+1,p\neq 2} \frac{p}{p-2}$, whose mean over the fifteen bases is 2.462. It gives $\hat\kappa = 0.759$. We stress that comparing this value with e^{γ} is meaningless: in the reparametrized model κ has a different scale, and such a "goodness-of-fit test" would only be testing whether the mean of C_b equals one. Since models I and C^{fix} are non-nested and have the same number of free parameters, the correct comparison is directly by likelihood: $\ln L = -51.53$ against -35.75, a difference of -15.78. The fixed correction is decisively rejected.

Logarithmic corrections. A substantive correction is needed here. The model $\lambda_b(t) = (\kappa/\ln b)(1 + c_1/t + c_2/t^2)$ used in the previous version is *not an admissible intensity*: at the estimates obtained there, $\hat{c}_1 = +0.108$, $\hat{c}_2 = -0.826$, the factor vanishes at $t = 0.8564$ and is negative to the left of that point, that is, for $n < 2.355$; in particular, at $t_0 = \ln 2$ it equals -0.5634. The bases with $n_{\min} = 2$ ($b = 2, 6, 10, 12, 18$) integrate the intensity over a region where it is negative, and the measure $\int \lambda\, dt$ is negative on that stretch. That the likelihood could be computed at all was a matter of chance: the smallest index that is an event is 3, that is, $t = 1.0986 > 0.8564$, so no negative values fell under the logarithm.

The correct parametrization of the same family is

$$\lambda_b(t) = \frac{\kappa}{\ln b} \exp\Big(\frac{c_1}{t} + \frac{c_2}{t^2}\Big), \tag{23}$$

which is identically positive and reduces to the null model at $c_1 = c_2 = 0$. It is estimated by the conditional point-process likelihood (14): the sum $\sum \ln \lambda(t_i)$ is taken over the events *excluding the last*, and the integral $\int \lambda$ runs from t_0 to the last event $t_0 + D_b$, not to the unknown frontier (Remark 8; the likelihood of the count alone is inadmissible here: it discards the positions of the events, which are the only thing that determines c_1, c_2). The result: $\hat{\kappa} = 1.748$, $\hat{c}_1 = 0.75$, $\hat{c}_2 = -1.44$; the likelihood ratio against $c_1 = c_2 = 0$ is 2.03 on two degrees of freedom, $p = 0.36$ (tag [S17]). Since $\hat{c}_1$ and $\hat{c}_2$ are strongly correlated, the χ^2_2 asymptotics are not guaranteed here, and we calibrated them by a parametric bootstrap of the conditional model: $p = 0.366$ from 10 000 replications, 95% quantile 5.93 against 5.99 for χ^2_2 (tag [S30]). There is no significant improvement. The estimate $\hat{c}_1$ is strongly correlated with $\hat{c}_2$ here and is not informative on its own: the one-parameter model (25) on the same data gives $\hat{c} = -0.40$ in the integral form and 1.03 with the exposure summed over the primes (§9).

9 The second-order term of the heuristic itself

So far we have tested the constant-intensity model (3). But the LPW heuristic itself predicts an intensity that is *not* constant at finite n, and this changes the meaning of comparing $\hat{\kappa}$ with e^γ.

Derivation. Let p be an odd prime, $p \nmid b - 1$, and let q be a prime divisor of $R_b(p)$. Then b has order p modulo q, whence $q \equiv 1 \pmod p$, and since $R_b(p)$ is odd, $q \equiv 1 \pmod{2p}$. Hence $R_b(p)$ has no prime divisors below $2p + 1$, and by Mertens' theorem the probability of primality is increased by the factor $\prod_{q<2p}(1 - 1/q)^{-1} \approx e^\gamma \ln(2p)$ compared with a random number of the same size. This is the derivation of [1, 3]:

$$\Pr\big(R_b(p) \text{ prime}\big) \approx \frac{e^\gamma \ln(a_b p)}{p \ln b}. \tag{24}$$

Wagstaff's refinement for $b = 2$ takes into account that $q \equiv \pm 1 \pmod 8$ and gives $a = 2$ for $p \equiv 3 \pmod 4$ and $a = 6$ for $p \equiv 1 \pmod 4$. In the continuous approximation, multiplying (24) by the density of primes $1/\ln n$ and passing to $t = \ln n$, we get

$$\lambda_b(t) = \frac{\kappa}{\ln b}\Big(1 + \frac{c_b}{t}\Big), \qquad c_b = \langle \ln a_b \rangle, \qquad \kappa = e^\gamma, \tag{25}$$

and, for the counting function, $\pi_b(x) \approx (e^\gamma/\ln b)\,(\ln x + c_b \ln\ln x)$. The asymptotics (2) are unaffected, but at $t \approx 10$–18 the correction c_b/t amounts to several percent—as much as the effects discussed in this paper.

Calibration of c_b. Wagstaff's device—replacing a_b by the smallest admissible divisor candidate—is crude, and we estimate c_b from the actual divisibility of $R_b(p)$ within the same Mertens heuristic. For a random number N, by Mertens,

$$\Pr(N \text{ prime} \mid \text{no prime divisors} < Y) \approx e^\gamma \ln Y / \ln N.$$

Hence $\Pr(R_b(p)\text{ prime}) \approx F_b(p,Y)\,e^\gamma \ln Y/(p\ln b)$, where $F_b(p,Y)$ is the indicator that $R_b(p)$ has no prime divisors below Y. Comparison with (24) gives

$$c_b(Y) = \big\langle F_b(p,Y)\ln Y - \ln p\big\rangle_p. \tag{26}$$

The indicator F_b is computed *exactly*. For $p > b$, $R_b(p)$ has a prime divisor below Y if and only if $b^p \equiv 1 \pmod q$ for some $q = 2kp+1 < Y$, prime or not. Indeed, the prime divisors of such a q that do not divide $b-1$ are congruent to 1 modulo p and divide $R_b(p)$; the part of q composed of divisors of $b-1$ divides $b-1<p$ by the lifting-the-exponent lemma and is congruent to 1 (mod p), hence equals 1. For 31 598 primes $p \in [2\cdot 10^4, 4\cdot 10^5)$ and $Y = 100p$, $1000p$, $3000p$ we obtain values of c_b that are stable in Y to within a standard error of 0.05 (script `cb_empirical.py`, tag [S19]; the values for different Y are obtained on the same set of p and are therefore correlated). At $Y = 1000p$ they range from 0.43 ($b=12$) to 1.11 ($b=20$), with mean 0.67. For ten bases out of fifteen the estimate of c_b is *below* $\ln 2 \approx 0.693$—the value given by a naive application of Mertens' theorem to the single condition $q \equiv 1 \pmod{2p}$; consequently, $\ln 2$ is not a lower bound for the effective c_b. Wagstaff's device (averaging $\ln a_b$ taking into account that b is a quadratic residue modulo q) gives systematically larger values, from 1.23 to 1.74 (tag [S14]); even for $b=2$ it gives $1.242 = \frac12(\ln 2 + \ln 6)$ against 0.96 ± 0.05. Moreover, this device is ambiguous: candidates divisible by a prime divisor of b are discarded, whereas those divisible by other small primes are not; thus for $b=18$ the residue condition coincides with the one for $b=2$, yet $c_{18} = 1.646$ against $c_2 = 1.242$. Below, the calibration of c_b by (26) is taken as the main one, and Wagstaff's variant is given only for comparison.

Where the values of c_b come from. The calibration can be predicted without computing $b^p \bmod q$ (script `cb_theory.py`, tag [S23]). A prime $q = 2kp+1$ divides $R_b(p)$ if and only if b lies in the subgroup of index $2k$ of $\mathbb{F}_q^*$. A necessary condition is that b be a quadratic residue modulo q, and this is determined by the Jacobi symbol (b/q). Within the subgroup of residues the natural heuristic is uniformity, which gives the probability $2/(2k) = 1/k$; for $k=1$ this equals 1, which is the classical case $2p+1 \mid M_p$ for $p \equiv 3 \pmod 4$. Hence the prediction

$$F_b^{\mathrm{th}}(p,Y) = \prod_{\substack{k<Y/(2p):\ q=2kp+1\ \text{prime},\\ (b/q)=+1}} \Big(1-\frac{1}{k}\Big), \qquad c_b^{\mathrm{th}} = \big\langle F_b^{\mathrm{th}}(p,Y)\ln Y - \ln p\big\rangle_p, \tag{27}$$

which uses only the primality of the numbers $2kp+1$ and the quadratic character of b. Table 10 compares it with the calibration on the same 31 598 primes p. The correlation is 0.976 and the mean difference is $+0.017$. A paired test over the individual p gives $\sum z_b^2 = 23.5$ with 15 degrees of freedom, $p = 0.074$. The prediction explains 95% of the between-base variation of c_b. The largest deviation is at $b=2$ ($+0.12$, $z=2.6$); its natural source is the biquadratic character of 2 (Gauss's criterion), which the model does not take into account. Equation (27) also implies a structural conclusion: in this approximation c_b depends only on the squarefree part of b. The pairs 2 and 18, 3 and 12, 5 and 20 have identical predictions, and the calibration confirms this within the errors. Thus the values of c_b—and the fact that Wagstaff's device overstates them—are explained by the distribution of primes of the form $2kp+1$ and by quadratic reciprocity, not by peculiarities of the data.

Conditional inference for a given c. By Remark 8, conditionally on D_b the interior events form a Poisson process with the same intensity on (t_0, t_0+D_b). But the heuristic (24) assigns a probability not on the t-axis but to each prime index, and the expected number of events in the window is a *sum over the primes*. Proposition 3 therefore carries over in the form

$$K \mid (D_b)_b \sim \mathrm{Poisson}\big(\kappa S_\Sigma(c)\big), \qquad S_\Sigma(c) = \sum_b \frac{1}{\ln b} \sum_{\substack{n_{\min}<p<n_{\max}\\ p\ \text{prime}}} \frac{\ln p + c_b}{p}, \tag{28}$$

Table 10. The second-order term c_b: calibration from the actual divisibility (equation (26), tag [S19]) and the prediction (27) from the primality of $2kp+1$ and the symbol (b/q) (tag [S23]); z is the paired z-statistic of the difference; 31 598 primes $p \in [2 \cdot 10^4,\ 4 \cdot 10^5)$, $Y = 1000p$.

b	2	3	5	6	7	10	11	12	13	14	15	17	18	19	20
calibration	0.96	0.44	1.06	0.59	0.49	0.55	0.47	0.43	0.67	0.53	0.54	0.76	0.86	0.54	1.11
prediction	0.84	0.46	1.01	0.50	0.51	0.52	0.53	0.46	0.75	0.51	0.52	0.71	0.84	0.54	1.01
z	+2.6	−0.4	+1.1	+2.0	−0.5	+0.7	−1.3	−0.8	−1.8	+0.4	+0.3	+1.0	+0.4	−0.1	+2.0

Table 11. Conditional inference with the second-order term: the estimate $\hat\kappa$ and the exact two-sided p against $\kappa = e^\gamma$. "c=0" is the constant-rate model; "meas., $\int$" are the measured c_b (tag [S19]) with the integral exposure (29) of the previous version; "meas., Σ" are the same c_b with the exposure (28) summed over the primes (the main variant); "W, Σ" is Wagstaff's device with the same sum. Truncation as in Table 7. Tags [S20], [S26].

		$\hat\kappa$				p		
truncation	K	c=0	meas., $\int$	meas., Σ	W, Σ	c=0	meas., Σ	W, Σ
none	203	1.838	1.620	1.780	1.618	0.672	1.000	0.178
$n > 10$	177	1.871	1.710	1.777	1.644	0.531	1.000	0.303
$n > 10^2$	141	1.830	1.703	1.724	1.615	0.771	0.740	0.261
$n > 10^3$	110	1.823	1.713	1.720	1.626	0.832	0.760	0.366
$n > 10^4$	75	1.789	1.690	1.693	1.614	1.000	0.712	0.429
$n > 10^4$, $N_b \geq 1$	75	1.757	1.660	1.663	1.585	0.966	0.598	0.340

where the inner sum omits the deterministically composite cases $p \mid b-1$ and $p = s$ (Aurifeuillean, §2.2). The estimator is $K/S_\Sigma(c)$, and the Garwood interval (15) applies with S replaced by $S_\Sigma(c)$. The Poisson law (28) is exact for a Poisson process with intensity (25). If the heuristic is read literally, as Bernoulli trials at the prime indices with probabilities (24) capped at one, conditioning on the last find remains valid (the trials before it are independent of that event), but the conditional law of K becomes Poisson-binomial: $\mathrm{Var}(K \mid D)/\mathbb{E}[K \mid D] = 0.934$ on the full sample and 0.973 at $n > 10$ (tag [S30]). The Poisson intervals and p-values are therefore conservative: the standard deviation is overstated by 3.5% and 1.4%. A crude allowance for the primality of the numbers $2p+1$, which is common to all bases, brings the variance-to-mean ratio back close to one (Remark 4). The continuous approximation (25) replaces this sum by an integral:

$$S(c) = \sum_b \frac{D_b + c_b \ln\bigl((t_0 + D_b)/t_0\bigr)}{\ln b}. \tag{29}$$

It is exact only asymptotically. By Mertens' first theorem $\sum_{p \le x} \ln p/p = \ln x + E + o(1)$ with $E = -1.3326\ldots$, and for origins $n_{\min} = 2$–19 the integral overstates the exposure of a base by 0.39–1.42 in t, 1.04 on average; under lower truncation $n > 10^2$ the difference is below one percent (tag [S26]). The previous version used form (29), which on the full sample and at $n > 10$ understated the estimate by 9.0% and 3.8% respectively; below, form (28) is taken as the main one.

Results. Table 11 and the simulation check yield five conclusions.

(i) Within the sequences the second-order term is only weakly determined, but it does not contradict the calibration. The profile MLE of a common c from the conditional likelihood with the exposure (28) on all the data is 1.03 with a 95% interval $[-0.18,\ 3.25]$: against $c = 0$ we obtain $p = 0.111$, against $c = \ln 2$, $p = 0.654$. At $n > 10$ the estimate is 0.82, close to the mean of the measured c_b (0.67), but the interval widens to $[-1.15,\ 5.19]$, and at $n > 10^3$ the information

practically vanishes (tag [S26]): a shift in c is compensated by a shift in κ. Therefore c_b has to be taken from outside—from theory or, as here, from the calibration (26). In the previous version the same profile estimate with the integral form (29) gave -0.40, and the interval $[-0.69, 0.44]$ excluded $c = \ln 2$ ($p = 0.017$, tag [S15]). That was an artifact of the approximation: near the lower end of the window the integral overstates the exposure, and the fit compensated with a negative c.

(ii) The inference about κ depends on the second-order term as strongly as on the observation scheme. Moving from the constant-intensity model to the measured c_b lowers the estimate by 3.2% on the full sample and by 5.0–5.8% under truncation (tag [S26]). This is comparable to the price of the choice of observation scheme (6.9%, §4.3) and not smaller than the residual gap between $(M - B)/S$ and e^γ (3.2% on the full sample, 5.1% at $n > 10$). Comparing the constant-intensity estimate with e^γ is not a test of LPW at finite n: LPW itself predicts that such an estimate lies *above* e^γ, by 3–8% (see (v)).

(iii) LPW with its own measured second-order term is consistent with the data. With the measured c_b the exact p lies within 0.598–1.000 at every truncation level. With Wagstaff's inflated values one obtains 0.26–0.43 at $n > 10$ and 0.178 on the full sample. The full sample includes $n \le 10$, where for small b the probability (24) exceeds one (at $b = 2$ for $p = 3$, 5 and 7), and the indices $2, 3, 5, 7$ themselves are dictated by the arithmetic of small numbers. We therefore regard the inference at $n > 10$ as the main one; with the sum over the primes it practically coincides with the full-sample inference (1.777 and 1.780).

(iv) The decrease under truncation is not explained by the second-order term. For $c = 0$ the estimate decreases as the lower bound grows, from 1.871 at $n > 10$ to 1.757 at $n > 10^4$ in the strictly consistent variant (-6.1%; with the selection $N_b \ge 2$, to 1.789, -4.4%). The previous version claimed that this decrease is predicted by (25) and disappears once it is taken into account: with the integral exposure (29) the four estimates at $n > 10$ lay within 1.690–1.713. We withdraw this claim: the levelling was produced by the same continuous approximation, which overstates the exposure most for small $n_{\min}$. With the sum over the primes the estimates are 1.777, 1.724, 1.720 and 1.693 (1.663 in the strictly consistent row $N_b \ge 1$); from $n > 10$ to $n > 10^4$ they decrease by 6.4% in the strictly consistent variant (4.7% with the selection $N_b \ge 2$)—as much as for $c = 0$. If the data are generated by the heuristic itself, the constant-intensity model overstates κ almost uniformly, by 5.7–8.3% at $n > 10$ (tag [S27]), so the heuristic predicts no decrease. The decrease itself is within the noise of nested subsamples: under scheme B a fall at least as large occurs with probability 0.229 (§5.4).

(v) The uncertainty of c_b and a check of the method on simulated data. The calibration error (± 0.045 per base) has practically no effect on the inference: its contribution to the standard deviation of $\hat\kappa$ is 0.003–0.004, against the Poisson 0.12. The joint 95% interval with the exposure (28)—a mixture of Garwood intervals over the scatter of c_b: the lower end solves $\langle \Pr(\mathrm{Gamma}(K, 1) \le L\, S_\Sigma(c)) \rangle_c = 0.025$, the upper end $\langle \Pr(\mathrm{Gamma}(K + 1, 1) \le U\, S_\Sigma(c)) \rangle_c = 0.975$—is [1.543, 2.042] on the full sample and [1.525, 2.059] at $n > 10$, and agrees to the third digit with the Garwood interval at fixed c_b: the calibration error changes the exposure by only 0.21% (tag [S26]). The previous version quoted a Jeffreys-type interval, [1.546, 2.036]; it is narrower than Garwood's, which is wrong for an interval that accounts for additional uncertainty. Using the prediction c_b^{th} instead of the calibration gives $\hat\kappa = 1.789$ and 1.784 (tag [S26]). The method was checked on data generated by the heuristic itself at the level of prime indices (tag [S27], 10 000 replications). Each prime $p > n_{\min}$ produced an event with probability (24) at $\kappa = e^\gamma$ and the measured c_b; the frontier was placed one mean gap beyond the actual last find, and the same analysis was applied. The estimator $K/S_\Sigma(c)$ is unbiased: $\mathbb{E}[\hat\kappa]/\kappa = 0.995$–$1.009$ under truncation up to $n > 10^3$, and the coverage of the Garwood interval is 0.955–0.961. On the same data the integral form (29) understates κ by 9% on the full sample (coverage 0.723) and by 3% at $n > 10$. The constant-intensity model overstates κ by 3–8%. At $n > 10^4$ both estimators with the second-order term are biased upward by 2.4–2.5%; this is the selection of

bases by $N_b \geq 2$ that the caption of Table 7 warns about. The earlier check (tag [S25]) generated data from the continuous intensity (25) and so could not detect the error of the integral form.

Consequence for the folklore. The term $c_b \ln\ln x$ in the counting function also explains the observation with which this work began: the normalized frequencies lie below $e^{-\gamma}$ and converge to it slowly from below. In the continuous approximation (25) the slope of t_k against k is $e^{-\gamma} \ln b/(1 + c_b\langle 1/t\rangle)$, so the normalized frequency is depressed by the factor $1 + c_b\langle 1/t\rangle$, where $\langle 1/t\rangle = \ln(t_N/t_0)/(t_N - t_0)$. With the origin fixed, this decays as $\ln t/t$, and the frequency tends to $e^{-\gamma}$ as $\ln\ln n/\ln n$—even more slowly than $1/\ln n$ (for $b = 2$: $\langle 1/t\rangle = 0.183$ against $1/t_N = 0.053$). In §3 we showed that fitting up to the right endpoint does not create the effect; here we see where it comes from: from the heuristic itself, not from $\kappa \neq e^{\gamma}$.

Summary. The question "are the data compatible with LPW?" splits into two: one about the leading term, $\kappa = e^{\gamma}$, and one about the second-order term, c_b. The latter is barely determined by the positions of the finds, but it can be calibrated from the actual divisibility of $R_b(p)$ (26). The expected number of finds must then be summed over the prime indices: at small n the integral with density $1/\ln n$ overstates the exposure by the Mertens constant. With the measured c_b the estimate is $\hat{\kappa} \approx 1.66$–$1.78$ at every lower truncation (0.1–7% below e^{γ}) with exact $p \geq 0.598$: LPW is not rejected. The decrease of the estimate under truncation is not removed by this term and is within the noise. The values of c_b themselves are predicted from first principles—from the primality of the numbers $2kp + 1$ and the quadratic character of b (27)—with correlation 0.976, and the method has been validated on simulated data. What remains open is a closed-form expression for c_b and the correction from higher power-residue characters (noticeable at $b = 2$).

10 Conclusions

1. The estimators of the normalized frequency of generalized repunit primes that are in common use are *not* biased by the fitting scheme: both the endpoint form and the OLS slope are exactly unbiased under scheme A and unbiased up to an error exponentially small in μ under scheme B (at most 0.73% for the endpoint form and 2.3% for OLS when $\mu \geq 7$). Only the reciprocal is biased, and that is a consequence of convexity. The common explanation of "slow convergence from below" in terms of censoring fails under both schemes; its actual explanation is the second-order term of the heuristic itself (item 7).

2. The genuine defect of current practice is that different forms of the estimator disagree by up to 29.5%, and that point values are published without confidence intervals even though the OLS standard errors in this problem are understated: the Newey–West correction increases them by a factor of 1.6, and the correct conditional null distribution is 5.6 times wider than the nominal one (§7). The classification of bases into "lucky" and "unlucky" flips when the form of the estimator is changed on unchanged data; for two of the fifteen bases even the sign of the deviation from $e^{-\gamma}$ flips. The largest discrepancy, for base 18, is created by a single anomalous gap after an origin fixed by arithmetic.

3. The main result is the exact conditional likelihood for scheme B. The unrecorded frontiers are Neyman–Scott nuisance parameters, and the profile estimator M/S is inconsistent. The position D_b of the last event is sufficient for the frontier, and conditionally on it $N_b - 1$ is distributed in the Poisson model as $\mathrm{Poisson}(\lambda_b D_b)$ under any rule for choosing the frontier that does not depend on the finds; the condition concerns the actual search horizon, while the published bound may depend on the finds (Remark 9). The same result is obtained by integrating over the frontier against the invariant measure dW_b. Hence $M - B \mid D \sim \mathrm{Poisson}(\kappa S)$: the correct pivot under a fixed frontier is Poisson rather than $\mathrm{Gamma}(M, 1)$, the correct pooled estimator is $(M - B)/S$ rather than $(M - 1)/S$, and

confidence intervals are available in closed form without simulation-based inversion. The conditional likelihood costs one event per base, not one event in total—an exact analogue of the loss of one degree of freedom per group in the Neyman–Scott example. The distinction between such schemes and the apparatus itself go back to reliability theory and to the theory of conditional likelihood; what is new here is the setting with an undocumented frontier, where the exposure cannot be taken from a protocol and has to be reconstructed from the finds (Remark 5).

4. Under (A1) this gives $\hat{\kappa} = 1.838$ with an exact 95% interval [1.594, 2.109] and $p = 0.672$ against e^{γ}. Under (A1) the reported ten-percent excess over LPW—including our own in an earlier version of this work—comes mainly from the observation scheme: moving to scheme B reduces it to 3.2%, and the remainder is explained by the second-order term of the heuristic itself (item 7). Under scheme B the old Gamma pivot rejects a true hypothesis in 11–17% of cases. Without (A1) the unknown stopping rule enters through a single parameter—the mean gap m between the last find and the frontier—and LPW is not rejected for any m from 0 to 3.2 (2.8 with the second-order term), that is, both under stopping at a find and under drought $g = 2$; only the residual excess depends on m, 10.8% at $m = 0$ and zero at $m = 1.42$. Moving to scheme B removes the upward trend of the pooled estimate under truncation in $n_{\min}$; both that trend and the remaining decline ($1.871 \to 1.757$ from $n > 10$ to $n > 10^4$; under scheme B a fall at least as large has probability 0.229) are within the noise of nested subsamples.

5. Besides the Poisson model and contiguous coverage of the search, the inference about the observation scheme rests on assumption (A1): the frontier was chosen without regard to the finds. It can practically not be checked from the finds themselves: $(M-1)/S$ and $(M-B)/S$ are functions of the same statistics and differ by 6.9%. Without (A1) only the upper bound M/S in the sense of expectations survives rigorously (Wald's identity); the rule "stop after a long stretch without finds" biases even $(M-B)/S$ upward, so there is no two-sided bracket, but such a violation creates no excess over LPW. The four documented frontiers do not settle (A1): three of them predate the data snapshot and give only lower bounds on the gap, while the synchronous $b = 2$ frontier is consistent with (A1). Under a deviation toward stopping close to a find the correct estimate lies, in the sense of expectations, between $(M-B)/S$ and M/S, and both bounds are consistent with LPW; the hybrid estimate with the known frontier for $b = 2$ is 1.846 (1.853 when $b = 10$ is added), and the estimate over the common window $n \le 2 \cdot 10^5$, covered with certainty, is 1.890. Gaps in the search bias the estimate downward; for LPW to be rejected from above there would have to be at least 23 of them. The practical consequence is the need to publish the search frontiers L_b (and, with them, the protocols of sieving completeness), as GIMPS does for $b = 2$. The example of base 5, where the probable primes found in 2024 lie beyond an unsearched gap, shows that this is not a formality.

6. The LPW conjecture is not rejected by the available data, and base-dependent corrections are not required. But both of these are statements about the low sensitivity of the tests: the conditional multinomial homogeneity test resolves only a spread of κ_b with a coefficient of variation of order 0.37, and the pooled test only an excess of the constant of at least 21.3% and a deficit of at least 19.2% (two-sided variant). Attaining a sensitivity of 10% would require the search frontiers to grow by a factor of about 10^{20}.

7. The LPW heuristic itself predicts at finite n an intensity $(\kappa/\ln b)(1 + c_b/t)$, so comparing the constant-intensity estimate with e^{γ} is not a test of LPW. The positions of the finds determine c_b only weakly, but it can be measured directly from the divisibility of $R_b(p)$: $c_b = 0.43$–1.11, with mean 0.67, whereas Wagstaff's device gives values twice as large. The exposure must then be summed over the prime indices: at small n the integral with

density $1/\ln n$ overstates it by the Mertens constant and would understate the estimate by 9.0%. With the measured c_b the estimate under (A1) is 1.66–1.78 at every truncation level, LPW is not rejected ($p \geq 0.598$), and the "slow convergence from below" is explained; the decrease of the estimate under truncation is not removed by this term. The c_b themselves are predicted from the primality of the numbers $2kp + 1$ and the quadratic character of b (correlation with the calibration 0.976) and depend only on the squarefree part of b; a simulation check on data generated by the heuristic at the level of prime indices confirms that the method is unbiased. A closed-form expression for c_b and the correction from higher power-residue characters remain open.

A Verification of the source data

All twenty sequences (fifteen of the main set and five from the check of the selection limit) were cross-checked against the OEIS entries on 27 August 2026. Since only the triples $(N_b, n_{\min}, n_{\max})$ enter the pooled analysis, Table 12 gives, for the main set, the number of known primes—from which N_b is obtained by subtracting one—and both extreme indices; the full lists of indices are contained in the script `analysis_v2.py`.

Table 12. Cross-check against OEIS. The column "last addition" gives the provenance of the largest known index; for $b = 2$ the last two exponents are taken from GIMPS.

b	OEIS	primes	$n_{\min}$	$n_{\max}$	last addition
2	A000043	52	2	136 279 841	GIMPS, Oct. 2024
3	A028491	23	3	8 530 117	R. Propper, 9 Nov. 2023
5	A004061	19	3	3 300 593	R. Propper, 2 Jan. 2022
6	A004062	17	2	3 360 347	R. Propper, 30 Oct. 2023
7	A004063	10	5	1 264 699	P. Bourdelais
10	A004023	11	2	8 177 207	Propper and Batalov, May 2021
11	A005808	13	17	1 868 983	P. Bourdelais
12	A004064	14	2	769 543	P. Bourdelais, Dec. 2014
13	A016054	13	5	1 503 503	P. Bourdelais
14	A006032	11	3	1 724 417	P. Bourdelais, 13 Jul. 2026
15	A006033	10	3	639 833	P. Bourdelais, 22 Apr. 2019
17	A006034	12	3	1 990 523	P. Bourdelais, 3 Aug. 2020
18	A133857	8	2	1 270 141	P. Bourdelais
19	A006035	11	19	209 359	P. Bourdelais, 27 Aug. 2010
20	A127995	9	3	984 349	P. Bourdelais

A separate warning to the reader who checks the data against secondary sources: summary tables (the "Repunit primes" page in the OEIS Wiki, the article "List of repunit primes" in Wikipedia) are updated with a delay and, at the time of writing, lagged the OEIS entries themselves by one term for the bases $b = 14, 15, 17, 18, 20$. The OEIS entries should be treated as authoritative.

Independent recomputation of the prefixes

Cross-checking against OEIS confirms that the listed indices give primes, but it does not guarantee *completeness*: a missing term would shift N_b, and with it both pooled estimates. We closed this gap by direct recomputation. For each of the twenty bases $2 \leq b \leq 26$ that are not perfect powers, an independent implementation[2] tested *all* indices $n < 10^4$ consecutively, not only those

[2] A hybrid CPU/GPU searcher: segmented sieving by small divisors, trial factoring in Montgomery arithmetic, PRP testing via GMP and GWNUM/IBDWT. The source code and the run logs are in the same repository, scripts `verify_sequences.sh` and `compare_verify.py`.

listed in OEIS. Negative verdicts are of two kinds: a divisor found (sieving and trial division; a certificate of compositeness) and a negative Fermat test. In the final run every verdict of the second kind was recomputed by a second, independent backend (a blanket double-check, as in GIMPS) so as to rule out a lost find.

The result: 153 terms confirmed, none spurious, none missing, no disagreement between backends—across all twenty bases. Coverage is uneven across bases (from one term at $b = 18$ to twenty-two at $b = 2$): the main set $b \leq 20$ accounts for 128 confirmed indices out of 233 known. Above 10^4 a complete recomputation is computationally out of reach, and there we still rely on OEIS.

We note that this recomputation was not a formality. The first run used a sampled double-check (5% of verdicts, the default), and the independent implementation declared $R_{23}(3181)$ composite, disagreeing with A204940; the disagreement was found by the comparison with OEIS, not by the double-check. Investigation traced the error to the implementation itself—a spike of FFT rounding error on the last iteration, after the last periodic check—and not to the OEIS data. After the fix (a rounding-error check after the loop and a fall-back to GMP when GWNUM fails) the recomputation was repeated with a blanket double-check, and no disagreements remained. We report this episode as an illustration of the thesis of §2 on completeness of coverage: a single negative verdict is unreliable, a false "composite" is invisible without recomputation, and that is exactly why frontiers and verification protocols should be published.

Declarations

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Competing interests

The author declares no competing interests.

Data and code availability

All sequences used are listed in Table 1.

Data. The script `fetch_data.sh` downloads the OEIS b-files for all fifteen sequences of the main set, and `fetch_extra.sh` does the same for the five bases $21 \leq b \leq 26$ used in the check of the selection limit (§2.2). The downloaded files are included in the repository, so the analysis can be reproduced without network access.

Analysis. Every number in this paper is marked with a tag of the form [S3], identifying the corresponding section or table, in the output of one of the scripts (`results_v2.txt` or `results_v3.txt`). The main script is `analysis_v2.py` (numpy, scipy, matplotlib). The pooled analysis, all confidence intervals, the homogeneity test, the power calculations and the structural models depend only on the triples $(N_b, n_{\min}, n_{\max})$; the full lists of indices are needed only for the $\hat{G}_b^{\mathrm{OLS}}$ column of Table 5, for §7, for the truncation and for the variable-intensity

models. The script reads them from the b-files in the `data/` directory (the loader is shared with `analysis.py` of version 1). The random number generator is initialized with a fixed seed (20260914), so all Monte Carlo quantities are reproducible bit for bit. The script also produces all four figures (directory `figs_v2/`, with English labels in `figs_v2/en/`). Table 3 (stopping rules that violate (A1)) is produced by a separate script, `stopping_rules.py` (tag [S13], seed 20260924). The second-order term (§9), model (23) and the homogeneity test at $B = 20$ are computed by `lpw_second_order.py` (tags [S14]–[S17], [S20]), the measurement of c_b by `cb_empirical.py` (tag [S19]; deterministic, about ten minutes of computation), its theoretical prediction by `cb_theory.py` (tag [S23]), the propagation of the uncertainty of c_b and the simulation check of the method by `sim_second_order.py` (tags [S24], [S25]), and the check of (A1) against the documented frontiers and the hybrid estimate by `frontiers.py` (tag [S18]; the sources of the frontiers are listed in the script itself). The exposure of the second-order term summed over the primes, the profile estimate of a common c, the simulation check with a generator at the level of prime indices and the simulation of the intercept of §7 are computed by `lpw_prime_sum.py` (tags [S26]–[S28], seed 20260930); the additional robustness checks (significance of the truncation profile, the common frontier $2 \cdot 10^5$, the choice of origin, base 18, power thresholds for a deficit, Figure 4) by `robust_extra.py` (tag [S29]); the computations requested in the editor's decision (the family $(M - Bm)/S$, the sensitivity to gaps, the Poisson-binomial law, the covariance of the bases, the selection on the number of finds, the bootstrap for model (23) and others) by `revision_v32.py` (tag [S30], seed 20261001). The output of these scripts is saved in `results_v3.txt`. The scripts `check_numbers_v3.py` and `check_all_numbers_v3.py` check all decimal numbers of both versions of the paper against it.

Independent recomputation of the sequences. `verify_sequences.sh` runs the searcher over all indices $n < 10^4$ for each base, and `compare_verify.py` compares the result with the OEIS b-files, returning a non-zero exit code on any disagreement (Appendix A).

A caveat on the Monte Carlo quantities. The main result (20), the intervals of Table 5, and Tables 6, 7 and 8 are *deterministic*: they are obtained by exact inversion of the Poisson distribution and do not depend on the random number generator. Only Tables 2, 4 and 3, the exact p-value of the homogeneity test and its power curve, and the null distributions of §7 depend on simulation; their accuracy is stated where they appear. This is an essential difference from the previous version, where the main interval estimate itself was simulation-based.

Repository: https://github.com/Wilps93/repunit-hunt, directory `paper/`. The file tree and the order of execution are described in `paper/README.md`.

References

[1] S. S. Wagstaff, Jr., *Divisors of Mersenne numbers*, Math. Comp. **40** (1983), no. 161, 385–397. DOI: 10.1090/S0025-5718-1983-0679454-X.

[2] C. Pomerance, *Recent developments in primality testing*, Math. Intelligencer **3** (1981), no. 3, 97–105. DOI: 10.1007/BF03022861.

[3] *Heuristics: deriving the Wagstaff Mersenne conjecture*, The PrimePages: prime number research and records. https://t5k.org/mersenne/heuristic.html (accessed 27 August 2026).

[4] H. Dubner, *Generalized repunit primes*, Math. Comp. **61** (1993), no. 204, 927–930. DOI: 10.1090/S0025-5718-1993-1185243-9.

[5] P. Bourdelais, *A generalized repunit conjecture*, NMBRTHRY mailing list, 25 June 2009. https://listserv.nodak.edu/cgi-bin/wa.exe?A2=NMBRTHRY;417ab0d6.0906 (accessed 25 August 2026).

[6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. https://oeis.org (accessed 27 August 2026).

[7] S. S. Wagstaff, Jr., *Two Mersenne prime conjectures*, J. Integer Seq. **28** (2025), article 25.7.2. https://cs.uwaterloo.ca/journals/JIS/VOL28/Wagstaff/wag14.html (accessed 24 September 2026).

[8] J. Dominguez, *Arithmetic bias in the distribution of Mersenne prime exponents and the divisor structure of* $p-1$, preprint, arXiv:2603.08994v6 (2026). https://arxiv.org/abs/2603.08994 (accessed 1 October 2026).

[9] J. Neyman and E. L. Scott, *Consistent estimates based on partially consistent observations*, Econometrica **16** (1948), no. 1, 1–32. DOI: 10.2307/1914288.

[10] E. B. Andersen, *Asymptotic properties of conditional maximum-likelihood estimators*, J. Roy. Statist. Soc. Ser. B **32** (1970), no. 2, 283–301. DOI: 10.1111/j.2517-6161.1970.tb00842.x.

[11] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika **23** (1976), no. 1, 4–9. DOI: 10.1112/S0025579300016442.

[12] H. Maier, *Primes in short intervals*, Michigan Math. J. **32** (1985), no. 2, 221–225. DOI: 10.1307/mmj/1029003189.

[13] A. Jha, *The Poisson tail conjecture for primes in short intervals*, preprint, arXiv:2605.23014 (2026). https://arxiv.org/abs/2605.23014 (accessed 27 August 2026).

[14] Great Internet Mersenne Prime Search (GIMPS), *PrimeNet*; Prime95/MPrime software, version 30.8. https://www.mersenne.org/ (accessed 27 August 2026; the *Milestones* report was read on 27 August 2026 at 04:30 UTC).

[15] B. Epstein, *Truncated life tests in the exponential case*, Ann. Math. Statist. **25** (1954), no. 3, 555–564. DOI: 10.1214/aoms/1177728723.

[16] B. Epstein and M. Sobel, *Some theorems relevant to life testing from an exponential distribution*, Ann. Math. Statist. **25** (1954), no. 2, 373–381. DOI: 10.1214/aoms/1177728793.

[17] D. J. Bartholomew, *The sampling distribution of an estimate arising in life testing*, Technometrics **5** (1963), no. 3, 361–374. DOI: 10.1080/00401706.1963.10490104.

[18] S. N. Sengupta, B. K. Sinha, and S. B. Bagchi, *Estimation of mean life and reliability for exponential life distribution using time censored sample data*, Statist. Probab. Lett. **79** (2009), no. 7, 899–906. DOI: 10.1016/j.spl.2008.11.026.

[19] M. Kivelä and M. A. Porter, *Estimating interevent time distributions from finite observation periods in communication networks*, Phys. Rev. E **92** (2015), no. 5, 052813. DOI: 10.1103/PhysRevE.92.052813.

[20] D. Strauss and P. M. Sadler, *Classical confidence intervals and Bayesian probability estimates for ends of local taxon ranges*, Math. Geol. **21** (1989), no. 4, 411–427. DOI: 10.1007/BF00897326.

[21] C. R. Marshall, *Confidence intervals on stratigraphic ranges*, Paleobiology **16** (1990), no. 1, 1–10. DOI: 10.1017/S0094837300009672.

[22] A. R. Solow, *Inferring extinction from sighting data*, Ecology **74** (1993), no. 3, 962–964. DOI: 10.2307/1940821.

[23] D. S. Robson and J. H. Whitlock, *Estimation of a truncation point*, Biometrika **51** (1964), no. 1–2, 33–39. DOI: 10.1093/biomet/51.1-2.33.

[24] Z. Liu, U. Bhandaram, and N. Garg, *Quantifying spatial under-reporting disparities in resident crowdsourcing*, Nat. Comput. Sci. **4** (2024), no. 1, 57–65. DOI: 10.1038/s43588-023-00572-6.

[25] T. Lancaster, *The incidental parameter problem since 1948*, J. Econometrics **95** (2000), no. 2, 391–413. DOI: 10.1016/S0304-4076(99)00044-5.

[26] J. D. Kalbfleisch and D. A. Sprott, *Application of likelihood methods to models involving large numbers of parameters* (with discussion), J. Roy. Statist. Soc. Ser. B **32** (1970), no. 2, 175–208. DOI: 10.1111/j.2517-6161.1970.tb00830.x.

[27] J. O. Berger, B. Liseo, and R. L. Wolpert, *Integrated likelihood methods for eliminating nuisance parameters*, Statist. Sci. **14** (1999), no. 1, 1–28. DOI: 10.1214/ss/1009211804.

[28] B. G. Lindsay, *Nuisance parameters, mixture models, and the efficiency of partial likelihood estimators*, Philos. Trans. Roy. Soc. London Ser. A **296** (1980), no. 1427, 639–662. DOI: 10.1098/rsta.1980.0197.

[29] T. S. Dokuchaev, *repunit-hunt: search for generalized repunit primes and statistical analysis*, GitHub repository, 2026. `https://github.com/Wilps93/repunit-hunt` (accessed 24 September 2026).

[30] F. Garwood, *Fiducial limits for the Poisson distribution*, Biometrika **28** (1936), no. 3–4, 437–442. DOI: 10.1093/biomet/28.3-4.437.